3.1 INTRODUCTION

Gate-level minimization is the design task of finding an optimal gate-level implementation of the Boolean functions describing a digital circuit. This task is well understood, but is difficult to execute by manual methods when the logic has more than a few inputs. Fortunately, this dilemma has been solved by computer-based logic synthesis tools that minimize a large set of Boolean equations efficiently and quickly. Nevertheless, it is important that a designer understands the underlying mathematical description and solution of the gate-level minimization problem. This chapter provides a foundation for your understanding of that important topic and will enable you to execute a manual design of simple circuits, preparing you for skilled use of modern design tools. The chapter will also introduce the role and use of hardware description languages in modern logic design methodology.

3.2 THE MAP METHOD

The complexity of the digital logic gates that implement a Boolean function is directly related to the complexity of the algebraic expression describing the function. Although the truth table representation of a function is unique, when it is expressed algebraically it can appear in many different, but equivalent, forms. Boolean expressions may be simplified by algebraic means as discussed in Section 2.4. However, this procedure of minimization is awkward, because it lacks specific rules to predict each succeeding step in the manipulative process. The map method presented in this section provides a simple, straightforward procedure for minimizing Boolean functions. This method may be regarded as a pictorial form of a truth table. The map method is also known as the *Karnaugh map* or *K-map* method.

A K-map is a diagram made up of squares, with each square representing one minterm of the function that is to be minimized. Since any Boolean function can be expressed as a sum of minterms, it follows that a Boolean function is recognized graphically in the map from the area enclosed by those squares whose minterms are included in the function. In fact, the map presents a visual diagram of all possible ways a function may be expressed in standard form. By recognizing various patterns, the user can derive alternative algebraic expressions for the same function, from which the simplest can be selected.

The simplified expressions produced by the map are always in one of the two standard forms: sum of products or product of sums. **It will be assumed that the simplest algebraic expression is one that has a minimum number of terms with the smallest possible number of literals in each term.** This expression produces a circuit diagram with a minimum number of gates and the minimum number of inputs to each gate. We will see subsequently that the simplest expression is not unique: It is sometimes possible to find two or more expressions that satisfy the minimization criteria. In that case, either solution is satisfactory.

Two-Variable K-Map

The two-variable K-map is shown in Fig. 3.1(a). There are four minterms for two variables; hence, the map consists of four squares, one for each minterm. The map is redrawn

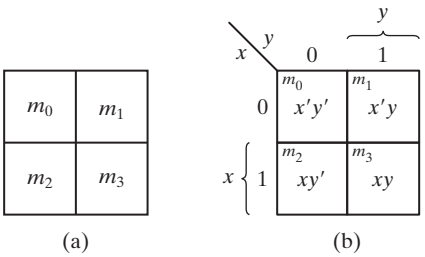


FIGURE 3.1
Two-variable K-map

in (b) to show the relationship between the squares and the two variables x and y. The 0 and 1 marked in each row and column designate the values of variables. Variable x appears primed in row 0 and unprimed in row 1. Similarly, y appears primed in column 0 and unprimed in column 1.

If we mark the squares whose minterms belong to a given function, the two-variable map becomes another useful way to represent any one of the 16 Boolean functions of two variables. As an example, the function xy is shown in Fig. 3.2(a). Since xy is equal to minterm m_3, a 1 is placed inside the square that belongs to m_3. Similarly, the function $x + y$ is represented in the map of Fig. 3.2(b) by three squares marked with 1's. These squares are found from the minterms of the function:

$$m_1 + m_2 + m_3 = x'y + xy' + xy = x + y$$

The three squares could also have been determined from the union of the squares of variable x in the second row and those of variable y in the second column, which encloses the area belonging to x or y. In each example, **the minterms at which the function is asserted are marked with a 1**.

Three-Variable K-Map

A three-variable K-map is shown in Fig. 3.3. There are eight minterms for three binary variables; therefore, the map consists of eight squares. Note that the minterms are

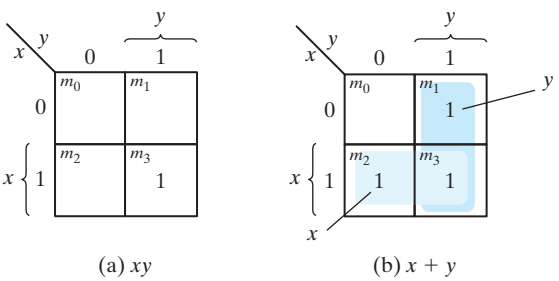


FIGURE 3.2
Representation of functions in the K-map

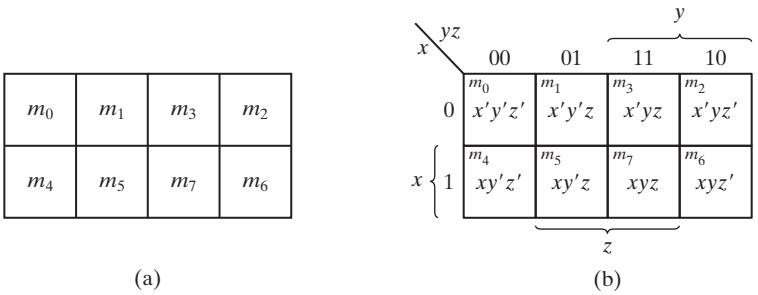


FIGURE 3.3
Three-variable K-map

arranged, not in a binary sequence, but in a sequence similar to the Gray code (Table 1.6). The characteristic of this sequence is that **only one bit changes in value from one adjacent column to the next.** The map drawn in part (b) is marked with numbered minterms in each row and each column to show the relationship between the squares and the three variables. For example, the square assigned to m_5 corresponds to row 1 and column 01. When these two numbers are concatenated, they give the binary number 101, whose decimal equivalent is 5. Each cell of the map corresponds to a unique minterm, so another way of looking at square $m_5 = xy'z$ is to consider it to be in the row marked x and the column belonging to $y'z$ (column 01). Note that there are four squares in which each variable is equal to 1 and four in which each is equal to 0. The variable appears unprimed in the former four squares and primed in the latter. For convenience, we write the variable with its letter symbol above or beside the four squares in which it is unprimed.

To understand the usefulness of the map in simplifying Boolean functions, we must recognize the basic property possessed by adjacent squares: **Any two adjacent squares in the map differ by only one variable,** which is primed in one square and unprimed in the other.[1] For example, m_5 and m_7 lie in two adjacent squares. Variable y is primed in m_5 and unprimed in m_7, whereas the other two variables are the same in both squares. From the postulates of Boolean algebra, it follows that the sum of two minterms in adjacent squares can be simplified to a single product term consisting of only two literals. To clarify this concept, consider the sum of two adjacent squares such as m_5 and m_7 :

$$m_5 + m_7 = xy'z + xyz = xz(y' + y) = xz$$

Here, the two squares differ by the variable y, which can be removed when the sum of the two minterms is formed. Thus, any two minterms in adjacent squares (vertically or horizontally, but not diagonally, adjacent) that are ORed together will cause a removal of the dissimilar variable. The next four examples explain the procedure for minimizing a Boolean function with a K-map.

[1] Squares that are neighbors on a diagonal are not considered to be adjacent.

EXAMPLE 3.1

Simplify the Boolean function

$$F(x, y, z) = \Sigma(2, 3, 4, 5)$$

First, a 1 is marked in each minterm square that represents the function. This is shown in Fig. 3.4, in which the squares for minterms 010, 011, 100, and 101 are marked with 1's. The next step is to find possible adjacent squares. These are indicated in the map by two shaded rectangles, each enclosing two 1's. The upper right rectangle represents the area enclosed by $x'y$. This area is determined by observing that the two-square area is in row 0, corresponding to x', and the last two columns, corresponding to y. Similarly, the lower left rectangle represents the product term xy'. (The second row represents x and the two left columns represent y'.) The sum of four minterms in the shaded squares can be replaced by a sum of only two product terms. The logical sum of these two product terms gives the simplified expression

$$F = x'y + xy'$$

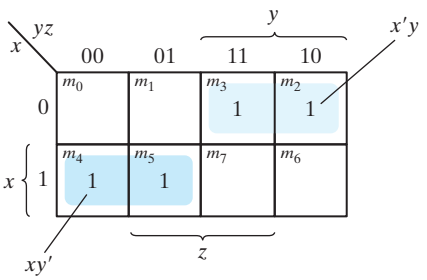


FIGURE 3.4
Map for Example 3.1, $F(x, y, z) = \Sigma(2, 3, 4, 5) = x'y + xy'$

■

In certain cases, two squares in the map are considered to be adjacent even though they do not touch each other. In Fig. 3.3(b), m_0 is adjacent to m_2 and m_4 is adjacent to m_6 because their minterms differ by one variable. This difference can be readily verified algebraically:

$$m_0 + m_2 = x'y'z' + x'yz' = x'z'(y' + y) = x'z'$$
$$m_4 + m_6 = xy'z' + xyz' = xz'(y' + y) = xz'$$

Consequently, we must modify the definition of adjacent squares to include this and other similar cases. We do so by considering the map as being drawn on a surface in which the right and left edges touch each other to form adjacent squares.

EXAMPLE 3.2

Simplify the Boolean function

$$F(x, y, z) = \Sigma(3, 4, 6, 7)$$

The map for this function is shown in Fig. 3.5. There are four squares marked with 1's, one for each minterm of the function. Two adjacent shaded squares in the third column are combined to give a two-literal term yz. The remaining two squares with 1's are also adjacent by the new definition. These two shaded squares, when combined, give the two-literal term xz'. The simplified function then becomes

$$F = yz + xz'$$

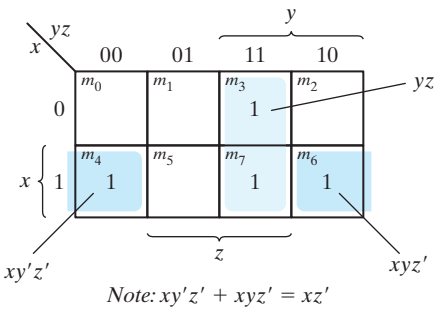


FIGURE 3.5
Map for Example 3.2, $F(x, y, z) = \Sigma(3, 4, 6, 7) = yz + xz'$

Consider now any combination of four adjacent squares in the three-variable map. Any such combination represents the logical sum of four minterms and results in an expression with only one literal. As an example, the logical sum of the four adjacent minterms 0, 2, 4, and 6 reduces to the single literal term z':

$$\begin{aligned} m_0 + m_2 + m_4 + m_6 &= x'y'z' + x'yz' + xy'z' + xyz' \\ &= x'z'(y' + y) + xz'(y' + y) \\ &= x'z' + xz' = (x' + x)z' = z' \end{aligned}$$

The number of adjacent squares that may be combined must always represent a number that is a power of two, such as 1, 2, 4, and 8. As more adjacent squares are combined, we obtain a product term with fewer literals.

One square represents one minterm, giving a term with three literals.

Two adjacent squares represent a term with two literals.

Four adjacent squares represent a term with one literal.

Eight adjacent squares encompass the entire three-variable map and produce a function that is always equal to 1.

EXAMPLE 3.3

Simplify the Boolean function

$$F(x, y, z) = \Sigma(0, 2, 4, 5, 6)$$

The map for F is shown in Fig. 3.6. First, we combine the four adjacent squares in the first and last columns to give the single literal term z'. The remaining single square, representing minterm 5, is combined with an adjacent square that has already been used once. This is not only permissible but also rather desirable, because the two adjacent squares give the two-literal term xy' and the single square represents the three-literal minterm $xy'z$. The simplified function is

$$F = z' + xy'$$

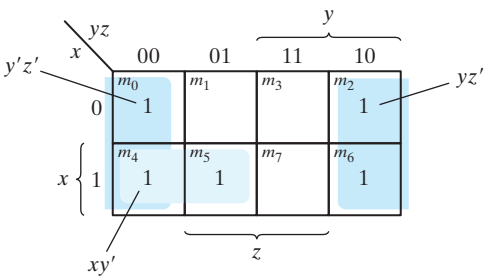


FIGURE 3.6
Map for Example 3.3, $F(x, y, z) = \Sigma(0, 2, 4, 5, 6) = z' + xy'$

If a function is not expressed in sum-of-minterms form, it is possible to use the map to obtain the minterms of the function and then simplify the function to an expression with a minimum number of terms. It is necessary, however, to *make sure that the algebraic expression is in sum-of-products form*. Each product term can be plotted in the map in one, two, or more squares. The minterms of the function are then read directly from the map.

EXAMPLE 3.4

For the Boolean function

$$F = A'C + A'B + AB'C + BC$$

(a) Express this function as a sum of minterms.
(b) Find the minimal sum-of-products expression.

Note that F is a sum of products, but not a sum of minterms. Three product terms in the expression have two literals and are represented in a three-variable map by two squares

each. The two squares corresponding to the first term, $A'C$, are found in Fig. 3.7 from the coincidence of A' (first row) and C (two middle columns) to give squares 001 and 011. Note that, in marking 1's in the squares, it is possible to find a 1 already placed there from a preceding term. This happens with the second term, $A'B$, which has 1's in squares 011 and 010. Square 011 is common with the first term, $A'C$, though, so only one 1 is marked in it. Continuing in this fashion, we determine that the term $AB'C$ belongs in square 101, corresponding to minterm 5, and the term BC has two 1's in squares 011 and 111. The function has a total of five minterms, as indicated by the five 1's in the map of Fig. 3.7. The minterms are read directly from the map to be 1, 2, 3, 5, and 7. The function can be re-expressed in sum-of-minterms form as

$$F(A, B, C) = \Sigma(1, 2, 3, 5, 7)$$

The sum-of-products expression, as originally given, has too many terms. It can be simplified, as indicated by the shaded squares in the map, to an expression with only two terms:

$$F = C + A'B$$

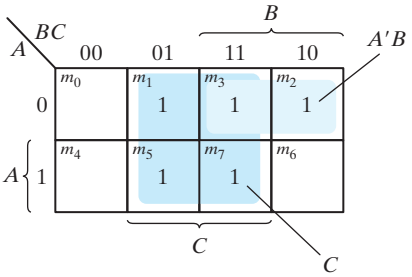


FIGURE 3.7
Map of Example 3.4, $A'C + A'B + AB'C + BC = C + A'B$

■

Practice Exercise 3.1

Simplify the Boolean function $F(x, y, z) = \Sigma(0, 1, 6, 7)$.

Answer: $F(x, y, z) = xy + x'y'$

Practice Exercise 3.2

Simplify the Boolean function $F(x, y, z) = \Sigma(0, 1, 2, 5)$.

Answer: $F(x, y, z) = x'z' + y'z$

Practice Exercise 3.3

Simplify the Boolean function $F(x, y, z) = \Sigma(0, 2, 3, 4, 6)$.

Answer: $F(x, y, z) = z' + x'y$

Practice Exercise 3.4

For the Boolean function $F(x, y, z) = xy'z + x'y + x'z + yz$, (a) express this function as a sum of minterms, and (b) find the minimal sum-of-products expression.

Answer: $F(x, y, z) = m1 + m2 + m3 + m5 + m7 = z + x'y = z + x'y$

3.3 FOUR-VARIABLE K-MAP

The map for Boolean functions of four binary variables (w, x, y, z) is shown in Fig. 3.8, which lists the 16 minterms and the squares assigned to each. In Fig. 3.8(b), the map is redrawn to show the relationship between the squares and the four variables. The rows and columns are numbered in a Gray code sequence, with only one digit changing value between two adjacent rows or columns. The minterm corresponding to each square can be obtained from the concatenation of the row number with the column number. For example, the numbers of the third row (11) and the second column (01), when concatenated, give the binary number 1101, the binary equivalent of decimal 13. Thus, the square in the third row and second column represents minterm m_{13}.

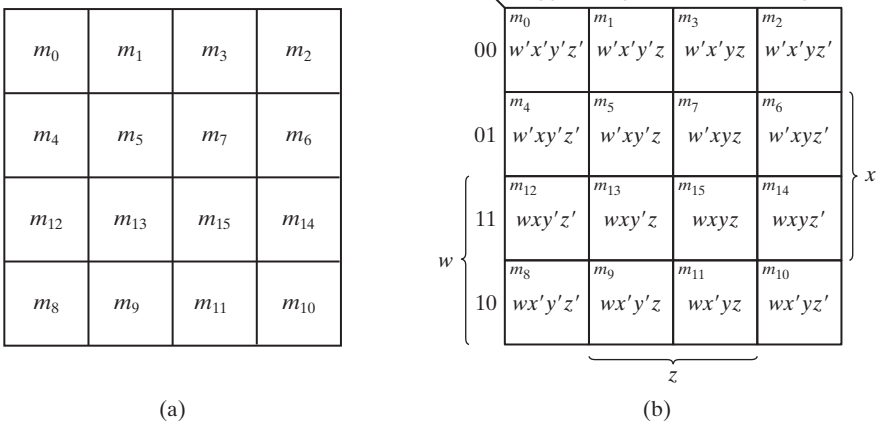


FIGURE 3.8
Four-variable map

The map minimization of four-variable Boolean functions is similar to the method used to minimize three-variable functions. Adjacent squares are defined to be squares next to each other (vertically or horizontally, but not diagonally). In addition, the map is considered to lie on a surface with the top and bottom edges, as well as the right and left edges, touching each other to form adjacent squares. For example, m_0 and m_2 form adjacent squares, as do m_3 and m_{11}. The combination of adjacent squares that is useful during the simplification process is easily determined from inspection of the four-variable map:

One square represents one minterm, giving a term with four literals.
Two adjacent squares represent a term with three literals.
Four adjacent squares represent a term with two literals.
Eight adjacent squares represent a term with one literal.
Sixteen adjacent squares produce a function that is always equal to 1.

No other combination of squares can simplify the function. The next two examples show the procedure used to simplify four-variable Boolean functions.

EXAMPLE 3.5

Simplify the Boolean function

$$F(w, x, y, z) = \Sigma(0, 1, 2, 4, 5, 6, 8, 9, 12, 13, 14)$$

Since the function has four variables, a four-variable map must be used. The minterms listed in the sum are marked by 1's in the map of Fig. 3.9. Eight shaded, adjacent squares marked with 1's can be combined to form the one literal term y'. The remaining three

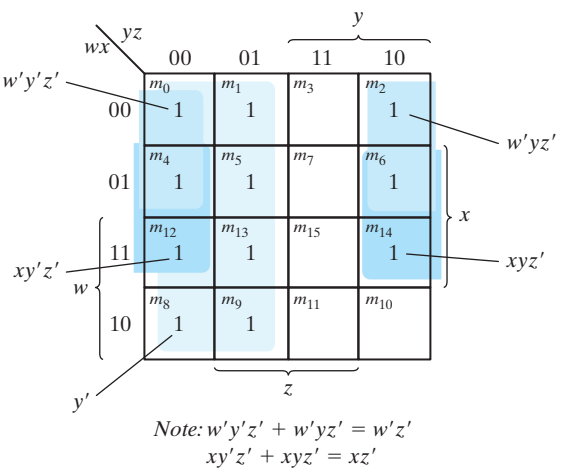


FIGURE 3.9
Map for Example 3.5, $F(w, x, y, z) = \Sigma(0, 1, 2, 4, 5, 6, 8, 9, 12, 13, 14) = y' + w'z' + xz'$

1's on the right cannot be combined to give a simplified term; they must be combined as two or four adjacent squares. The larger the number of squares combined is, the smaller will be the number of literals in the term. In this example, the top two 1's on the right are combined with the top two 1's on the left to give the term $w'z'$. Note that it is permissible to use the same square more than once. We are now left with a square marked by 1 in the third row and fourth column (square 1110). Instead of taking this square alone (which will give a term with four literals), we combine it with squares already used to form an area of four adjacent squares. These squares make up the two middle rows and the two end columns, giving the term xz'. The simplified function is

$$F = y' + w'z' + xz'$$

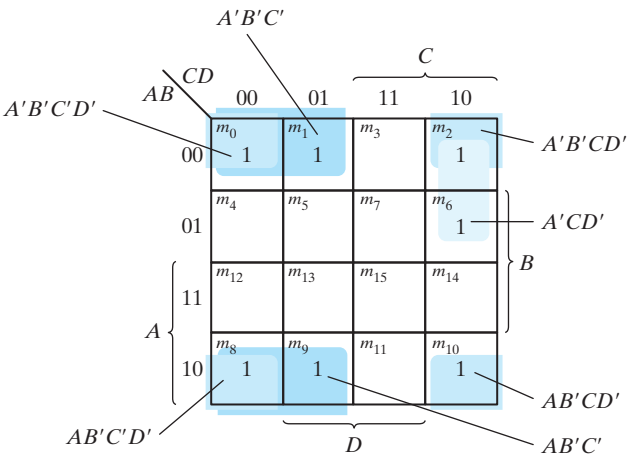
The number of terms and the number of literals has been reduced.

EXAMPLE 3.6

Simplify the Boolean function

$$F = A'B'C' + B'CD' + A'BCD' + AB'C'$$

The area in the map covered by this function consists of the squares marked with 1's in Fig. 3.10. The function has four variables and, as expressed, consists of three terms with three literals each and one term with four literals. Each term with three literals is represented in the map by two squares. For example, $A'B'C'$ is represented in squares



Note: $A'B'C'D' + A'B'CD' = A'B'D'$
$AB'C'D' + AB'CD' = AB'D'$
$A'B'D' + AB'D' = B'D'$
$A'B'C' + AB'C' = B'C'$

FIGURE 3.10
Map for Example 3.6, $A'B'C' + B'CD' + A'BCD' + AB'C = B'D' + B'C' + A'CD'$

0000 and 0001. The function can be simplified in the map by taking the 1's in the four corners to give the term $B'D'$. This is possible because these four squares are adjacent when the map is drawn in a surface with top and bottom edges, as well as left and right edges, touching one another. The two left-hand 1's in the top row are combined with the two 1's in the bottom row to give the term $B'C'$. The remaining 1 may be combined in a two-square area to give the term $A'CD'$. The simplified function has fewer terms, with fewer literals:

$$F = B'D' + B'C' + A'CD'$$

■

Practice Exercise 3.5

Simplify the Boolean function $F(w, x, y, z) = \Sigma(0, 1, 3, 8, 9, 10, 11, 12, 13, 14, 15)$.

Answer: $F(w, x, y, z) = x'y' + x'z$

Practice Exercise 3.6

Simplify the Boolean function $F(w, x, y, z) = \Sigma\ (0, 2, 4, 6, 8, 10, 11)$.

Answer: $F(w, x, y, z) = w'z' + x'z' + wx'y$

Prime Implicants

In choosing adjacent squares in a map, we must ensure that (1) all the minterms of the function are covered when we combine the squares, (2) the number of terms in the expression is minimized, and (3) there are no redundant terms (i.e., minterms already covered by other terms). Sometimes there may be two or more expressions that satisfy the simplification criteria. The procedure for combining squares in the map may be made more systematic if we understand the meaning of two special types of terms. We have seen that a product term is an implicant of the function to which it belongs. **A *prime implicant* is a product term obtained by combining the maximum possible number of adjacent squares in the map.** Thus, an implicant is prime if no other implicant having fewer literals covers it. If a minterm in a square is covered by only one prime implicant, that prime implicant is said to be *essential, that is, it cannot be removed from a description of the function.*

The prime implicants of a function can be obtained from the map by combining all possible maximum numbers of squares. This means that a single 1 on a K-map represents a prime implicant if it is not adjacent to any other 1's. Two adjacent 1's form a prime implicant, provided that they are not within a group of four adjacent squares. Four adjacent 1's form a prime implicant if they are not within a group of eight adjacent squares, and so on. The essential prime implicants are found by looking at each square marked with a 1 and checking the number of prime implicants that cover it. *A prime implicant is essential if it is the only prime implicant that covers the minterm.*

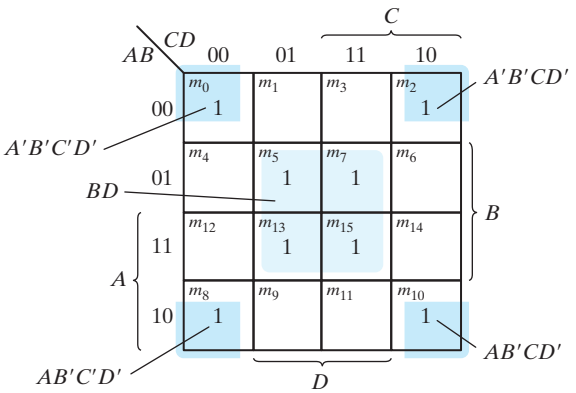
Consider the following four-variable Boolean function:

$$F(A, B, C, D) = \Sigma(0, 2, 3, 5, 7, 8, 9, 10, 11, 13, 15)$$

Some of the minterms of the function are marked with 1's in the maps of Fig. 3.11—we have omitted m_3, m_9, and m_{11}. The partial map (Fig. 3.11(a)) shows two essential prime implicants, each formed by collapsing four cells into a term having only two literals. One term is essential because there is only one way to include minterm m_0 within four adjacent squares. These four squares define the term $B'D'$. Similarly, there is only one way that minterm m_5 can be combined with four adjacent squares, and this gives the second term BD. The two essential prime implicants cover eight minterms. The three minterms that were omitted from the partial map (m_3, m_9, and m_{11}) must be considered next.

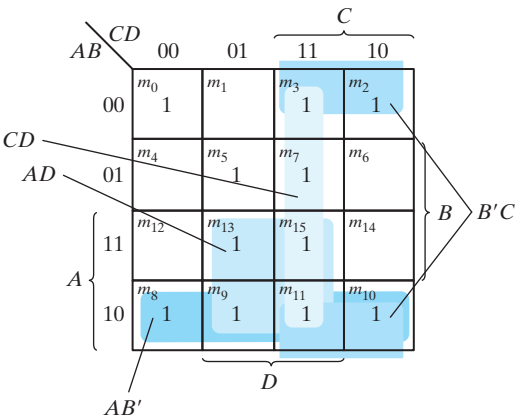
Figure 3.11(b) shows all possible ways that the three minterms can be covered with prime implicants. Minterm m_3 can be covered with either prime implicant CD or prime implicant $B'C$. Minterm m_9 can be covered with either AD or AB'. Minterm m_{11} is covered with any one of the four prime implicants. The simplified expression is obtained from the logical sum of the two essential prime implicants and any two prime implicants that cover minterms m_3, m_9, and m_{11}. There are four possible ways that the function can be expressed with four product terms of two literals each:

$$\begin{aligned} F &= BD + B'D' + CD + AD \\ &= BD + B'D' + CD + AB' \\ &= BD + B'D' + B'C + AD \\ &= BD + B'D' + B'C + AB' \end{aligned}$$



Note: $A'B'C'D' + A'B'CD' = A'B'D'$
$AB'C'D' + AB'CD' = AB'D'$
$A'B'D' + AB'D' = B'D'$

(a) Essential prime implicants BD and $B'D'$



(b) Prime implicants CD, $B'C$, AD, and AB'

FIGURE 3.11
Simplification using prime implicants

The previous example has demonstrated that the identification of the prime implicants in the map helps in determining the alternatives that are available for obtaining a simplified expression.

The procedure for finding the simplified expression from the map requires that we first determine all the essential prime implicants. **The simplified expression is obtained from the logical sum of all the essential prime implicants, plus other prime implicants that may be needed to cover any remaining minterms not covered by the essential prime implicants.** Occasionally, there may be more than one way of combining squares, and each combination may produce an equally simplified expression.

Practice Exercise 3.7

Find the prime implicants of the Boolean function $F(w, x, y, z) = \Sigma(0, 2, 4, 5, 6, 7, 8, 10, 13, 14, 15)$.

Answer: $x'z'$, xz, xy, $w'x$

Five-Variable K-Map

Maps for more than four variables are not as simple to use as maps for four or fewer variables. A five-variable map needs 32 squares and a six-variable map needs 64 squares. When the number of variables becomes large, the number of squares becomes excessive and the geometry for combining adjacent squares becomes more involved.

Maps for more than four variables are difficult to use and will not be considered here.

3.4 PRODUCT-OF-SUMS SIMPLIFICATION

The minimized Boolean functions derived from the map in all previous examples were expressed in sum-of-products form. With a minor modification, the product-of-sums form can be obtained.

The procedure for obtaining a minimized function in product-of-sums form follows from the basic properties of Boolean functions. The 1's placed in the squares of the map represent the minterms of the function. The minterms not included in the standard sum-of-products form of a function denote the complement of the function. From this observation, we see that the complement of a function is represented in the map by the squares not marked by 1's. If we mark the empty squares by 0's and combine them into valid adjacent squares, we obtain a simplified sum-of-products expression of the complement of the function (i.e., of F'). The complement of F' gives us back the function F in product-of-sums form (a consequence of DeMorgan's theorem). Because of the generalized DeMorgan's theorem, the function so obtained is automatically in product-of-sums form. We will show this is by example.

EXAMPLE 3.7

Simplify the following Boolean function into (a) sum-of-products form and (b) product-of-sums form:

$$F(A, B, C, D) = \Sigma(0, 1, 2, 5, 8, 9, 10)$$

The 1's marked in the map of Fig. 3.12 represent all the minterms of the function. The squares marked with 0's represent the minterms not included in F and therefore denote the complement of F. Combining the squares with 1's gives the simplified function in sum-of-products form:

(a) $F = B'D' + B'C' + A'C'D$

If the squares marked with 0's are combined, as shown in the diagram, we obtain the simplified complemented function:

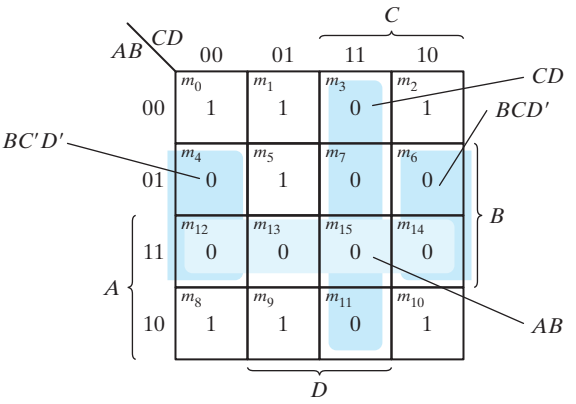
$$F' = AB + CD + BD'$$

Applying DeMorgan's theorem (by taking the dual and complementing each literal as described in Section 2.4), we obtain the simplified function in product-of-sums form:

(b) $F = (A' + B')(C' + D')(B' + D)$

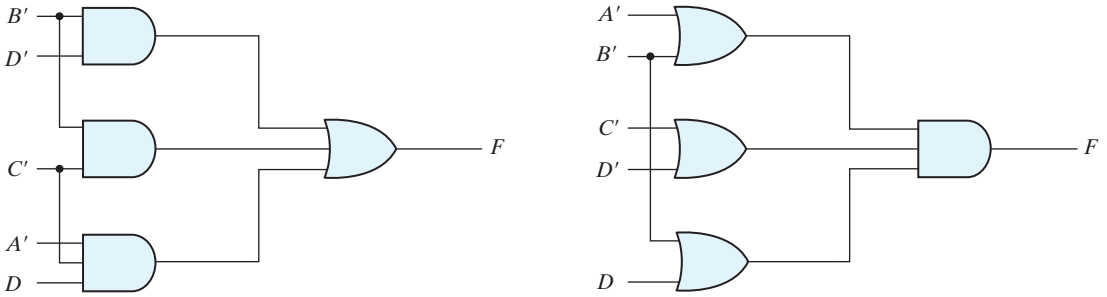
■

The gate-level implementation of the simplified expressions obtained in Example 3.7 is shown in Fig. 3.13. The sum-of-products expression is implemented in (a) with a group of AND gates, one for each AND term. The outputs of the AND gates are connected to



Note: $BC'D' + BCD' = BD'$

FIGURE 3.12
Map for Example 3.7, $F(A, B, C, D) = \Sigma(0, 1, 2, 5, 8, 9, 10) = BD + BC + ACD = (A' + B')(C' + D')(B' + D)$



(a) $F = B'D' + B'C' + A'C'D$ (b) $F = (A' + B')(C' + D')(B' + D)$

FIGURE 3.13
Gate implementations of the function of Example 3.7

the inputs of a single OR gate. The same function is implemented in (b) in its product-of-sums form with a group of OR gates, one for each OR term. The outputs of the OR gates are connected to the inputs of a single AND gate. In each case, it is assumed that the input variables are directly available in their complement, so inverters are not needed. The configuration pattern established in Fig. 3.13 is the general form by which any Boolean function is implemented when expressed in one of the standard forms. AND gates are connected to a single OR gate when in sum-of-products form; OR gates are connected to a single AND gate when in product-of-sums form. Either configuration forms two levels of gates. Thus, the implementation of a function in a standard form is said to be a two-level implementation. The two-level implementation may not be practical, depending on the number of inputs to the gates.

Example 3.7 showed the procedure for obtaining the product-of-sums simplification when the function is originally expressed in the sum-of-minterms canonical form. The procedure is also valid when the function is originally expressed in the product-of-maxterms canonical form. Consider, for example, the truth table that defines the function F in Table 3.1. In sum-of-minterms form, this function is expressed as

$$F(x, y, z) = \Sigma(1, 3, 4, 6)$$

Table 3.1
Truth Table of Function F

x	y	z	F
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

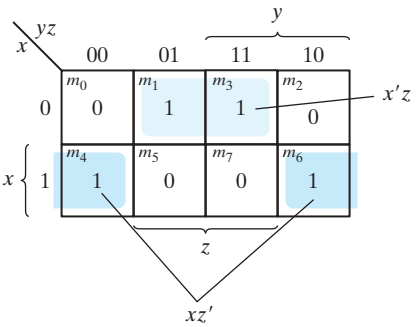


FIGURE 3.14
Map for the function of Table 3.1

In product-of-maxterms form, it is expressed as

$$F(x, y, z) = \Pi(0, 2, 5, 7)$$

In other words, the 1's of the function represent the minterms and the 0's represent the maxterms. The map for this function is shown in Fig. 3.14. One can start simplifying the function by first marking the 1's for each minterm that the function is a 1. The remaining squares are marked by 0's. If, instead, the product of maxterms is initially given, one can start marking 0's in those squares listed in the function; the remaining squares are then marked by 1's. Once the 1's and 0's are marked, the function can be simplified in either one of the standard forms. For the sum of products, we combine the 1's to obtain

$$F = x'z + xz'$$

For the product of sums, we combine the 0's to obtain the simplified complemented function

$$F' = xz + x'z'$$

which shows that the exclusive-OR function is the complement of the equivalence function (Section 2.7). Taking the complement of F', we obtain the simplified function in product-of-sums form:

$$F = (x' + z')(x + z)$$

To enter a function expressed in product-of-sums form into the map, use the complement of the function to find the squares that are to be marked by 0's. For example, the function

$$F = (A' + B' + C')(B + D)$$

can be entered into the map by first taking its complement, namely,

$$F' = ABC + B'D'$$

and then marking 0's in the squares representing the minterms of F'. The remaining squares are marked with 1's.

Practice Exercise 3.8

Simplify the Boolean function $F(w, x, y, z) = \Sigma(0, 2, 8, 10, 12, 13, 14)$ into (a) sum-of-products form and (b) product-of-sums form. Derive the truth table of F.

Answer: $F(w, x, y, z) = x'z' + wz' + wxy'$

$F'(w, x, y, z) = w'x + yz + x'z$

$F(w, x, y, z) = (w + x')(y' + z')(x + z')$

wxyz F	wxyz F
0000 1	1000 1
0001 0	1001 0
0010 1	1010 1
0011 0	1011 0
0100 0	1100 1
0101 0	1101 1
0110 0	1110 1
0111 0	1111 0

3.5 DON'T-CARE CONDITIONS

The logical sum of the minterms associated with a Boolean function specifies the conditions under which the function is equal to 1. The function is equal to 0 for the rest of the minterms. This pair of conditions assumes that all the combinations of the values for the variables of the function are valid. In practice, in some applications the function is not specified for certain combinations of the variables. As an example, the four-bit binary code for the decimal digits has six combinations that are not used and consequently are considered to be unspecified. Functions that have unspecified outputs for some input combinations are called *incompletely specified functions*. In most applications, we simply don't care what value is assumed by the function for the unspecified minterms. For this reason, it is customary to call the unspecified minterms of a function *don't-care conditions*. These don't-care conditions can be used on a map to provide further simplification of the Boolean expression.[2]

A don't-care minterm is a combination of variables whose logical value is not specified. Such a minterm cannot be marked with a 1 in the map, because it would require that

[2]The Quine–McCluskey method uses a tabular format as an alternative to the Karnaugh map methods used in our examples. As such, it is suitable for implementation in software.

the function always be a 1 for such a combination. Likewise, putting a 0 on the square requires the function to be 0. To distinguish the don't-care condition from 1's and 0's, an X is used. Thus, an X inside a square in the map indicates that we don't care whether the value of 0 or 1 is assigned to F for the particular minterm.

In choosing adjacent squares to simplify the function in a map, the don't-care minterms may be assumed to be either 0 or 1. When simplifying the function, we can choose to include each don't-care minterm with either the 1's or the 0's, depending on which combination gives the simplest expression.

EXAMPLE 3.8

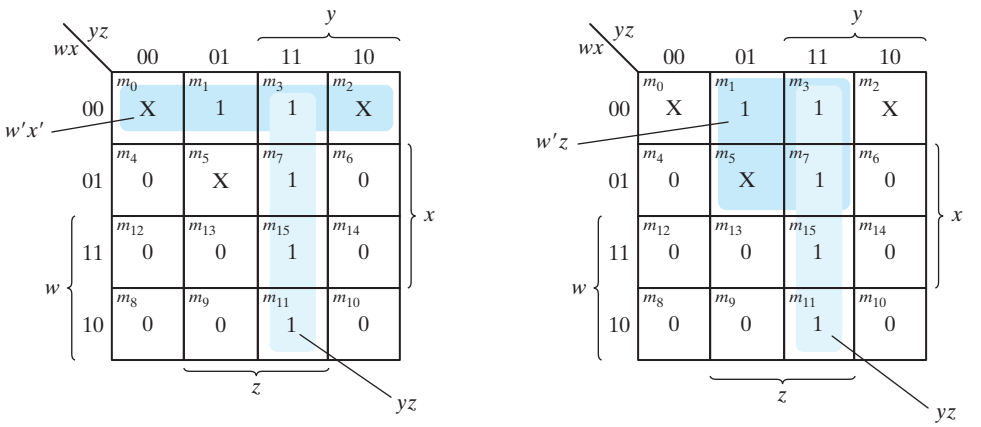
Simplify the Boolean function

$$F(w, x, y, z) = \Sigma(1, 3, 7, 11, 15)$$

which has the don't-care conditions

$$d\,(w, x, y, z) = \Sigma(0, 2, 5)$$

The minterms of F are the variable combinations that make the function equal to 1. The minterms of d are the don't-care minterms that may be assigned either 0 or 1. The map simplification is shown in Fig. 3.15. The minterms of F are marked by 1's, those of d are marked by X's, and the remaining squares are filled with 0's. To get the simplified expression in sum-of-products form, we must include all five 1's in the map, but we may or may not include any of the X's, depending on the way the function is simplified. The term yz covers the four minterms in the third column. The remaining minterm, m_1, can be combined with minterm m_3 to give the three-literal term $w'x'z$. However, by including one or two adjacent X's we can combine four adjacent squares to give a two-literal



(a) $F = yz + w'x'$ (b) $F = yz + w'z$

FIGURE 3.15
Example with don't-care conditions

term. In Fig. 3.15(a), don't-care minterms 0 and 2 are included with the 1's, resulting in the simplified function

$$F = yz + w'x'$$

In Fig. 3.15(b), don't-care minterm 5 is included with the 1's, and the simplified function is now

$$F = yz + w'z$$

Either one of the preceding two expressions satisfies the conditions stated for this example. ■

The previous example has shown that the don't-care minterms in the map are initially marked with X's and are considered as being either 0 or 1. The choice between 0 and 1 is made depending on the way the incompletely specified function is simplified. Once the choice is made, the simplified function obtained will consist of a sum of minterms that includes those minterms, which were initially unspecified and have been chosen to be included with the 1's. Consider the two simplified expressions obtained in Example 3.8:

$$F(w, x, y, z) = yz + w'x' = \Sigma(0,1,2,3,7,11,15)$$
$$F(w, x, y, z) = yz + w'z = \Sigma(1,3,5,7,11,15)$$

Both expressions include minterms 1, 3, 7, 11, and 15 that make the function F equal to 1. The don't-care minterms 0, 2, and 5 are treated differently in each expression. The first expression includes minterms 0 and 2 with the 1's and leaves minterm 5 with the 0's. The second expression includes minterm 5 with the 1's and leaves minterms 0 and 2 with the 0's. The two expressions represent two functions that are not algebraically equal. Both cover the specified minterms of the function, but each covers different don't-care minterms. As far as the incompletely specified function is concerned, either expression is acceptable because the only difference is in the value of F for the don't-care minterms.

It is also possible to obtain a simplified product-of-sums expression for the function of Fig. 3.15. In this case, the only way to combine the 0's is to include don't-care minterms 0 and 2 with the 0's to give a simplified complemented function:

$$F' = z' + wy'$$

Taking the complement of F' gives the simplified expression in product-of-sums form:

$$F(w, x, y, z) = z(w' + y) = \Sigma(1,3,5,7,11,15)$$

In this case, we include minterms 0 and 2 with the 0's and minterm 5 with the 1's.

Practice Exercise 3.9

Simplify the Boolean function $F(w, x, y, z) = \Sigma(4, 5, 6, 7, 12)$ with don't-care function $d(w, x, y, z) = \Sigma(0, 8, 13)$.

Answer: $F(w, x, y, z) = xy' + xw'$

3.6 NAND AND NOR IMPLEMENTATION

Digital circuits are frequently constructed with NAND or NOR gates rather than with AND and OR gates. NAND and NOR gates are easier to fabricate with electronic components and are the basic gates used in all IC digital logic families. Because of the prominence of NAND and NOR gates in the design of digital circuits, rules and procedures have been developed for the conversion from Boolean functions given in terms of AND, OR, and NOT into equivalent NAND and NOR logic diagrams.

NAND Circuits

The NAND gate is said to be a *universal* gate because any logic circuit can be implemented with it. To show that any Boolean function can be implemented with NAND gates, we need only to show that the logical operations of AND, OR, and complement can be obtained with NAND gates alone. This is indeed shown in Fig. 3.16. The complement operation is obtained from a one-input NAND gate that behaves exactly like an inverter. The AND operation requires two NAND gates. The first produces the NAND operation and the second inverts the logical sense of the signal. The OR operation is achieved through a NAND gate with additional inverters in each input.

A convenient way to implement a Boolean function with NAND gates is to obtain the simplified Boolean function in terms of Boolean operators and then convert the function to NAND logic. The conversion of an algebraic expression from AND, OR, and complement to NAND can be done by simple circuit manipulation techniques that change AND–OR diagrams to NAND diagrams.

To facilitate the conversion to NAND logic, it is convenient to define an alternative graphic symbol for the gate. Two equivalent graphic symbols for the NAND gate are shown in Fig. 3.17. The AND-invert symbol has been defined previously and consists of an AND graphic symbol followed by a small circle negation indicator referred to as a bubble. Alternatively, it is possible to represent a NAND gate by an OR graphic symbol that is preceded by a bubble in each input. The invert-OR symbol for the NAND gate

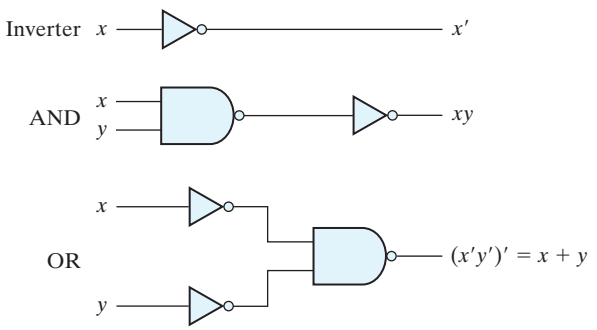
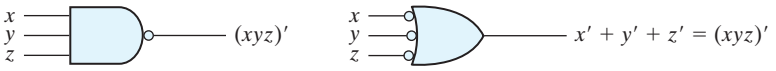


FIGURE 3.16
Logic operations with NAND gates



(a) AND-invert (b) Invert-OR

FIGURE 3.17
Two graphic symbols for a three-input NAND gate

follows DeMorgan's theorem and *the convention that the negation indicator (bubble) denotes complementation*. The two graphic symbols' representations are useful in the analysis and design of NAND circuits. When both symbols are mixed in the same diagram, the circuit is said to be in mixed notation.

Two-Level Implementation

The implementation of two-level Boolean functions with NAND gates requires that the functions be in sum-of-products form. To see the relationship between a sum-of-products expression and its equivalent NAND implementation, consider the logic diagrams drawn in Fig. 3.18. All three diagrams are equivalent and implement the function

$$F = AB + CD$$

The function is implemented in Fig. 3.18(a) with AND and OR gates. In Fig. 3.18(b), the AND gates are replaced by NAND gates and the OR gate is replaced by a NAND gate with an invert-OR graphic symbol. Remember that a bubble denotes complementation and two bubbles along the same line represent double complementation, so both can be

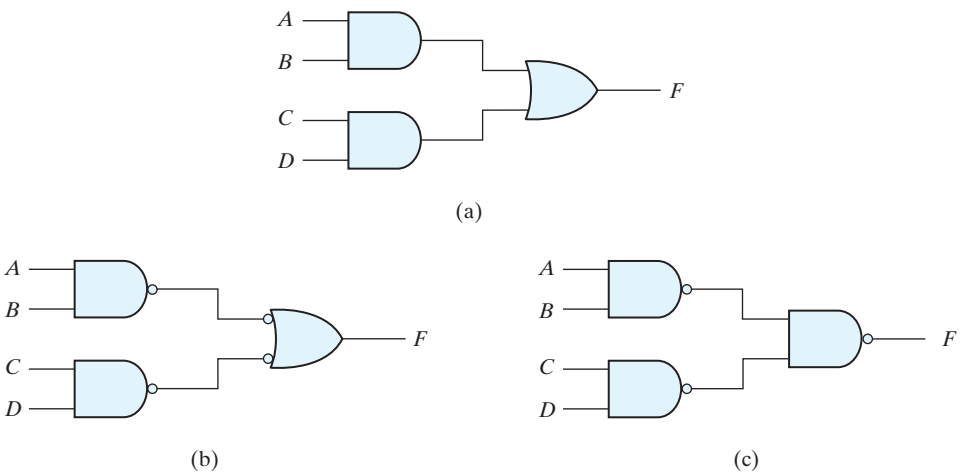


FIGURE 3.18
Three ways to implement $F = AB + CD$

removed. Removing the bubbles on the gates of (b) produces the circuit of (a). Therefore, the two diagrams implement the same function and are equivalent.

In Fig. 3.18(c), the output NAND gate is redrawn with the AND-invert graphic symbol. In drawing NAND logic diagrams, the circuit shown in either Fig. 3.18(b) or (c) is acceptable. The one in Fig. 3.18(b) is in mixed notation and represents a more direct relationship to the Boolean expression it implements. The NAND implementation in Fig. 3.18(c) can be verified algebraically. The function it implements can easily be converted to sum-of-products form by DeMorgan's theorem:

$$F = ((AB)'(CD)')' = AB + CD$$

EXAMPLE 3.9

Implement the following Boolean function with NAND gates:

$$F(x, y, z) = (1, 2, 3, 4, 5, 7)$$

The first step is to simplify the function into sum-of-products form. This is done by means of the map of Fig. 3.19(a), from which the simplified function is obtained:

$$F = xy' + x'y + z$$

The two-level NAND implementation is shown in Fig. 3.19(b) in mixed notation. Note that input z must have a one-input NAND gate (an inverter) to compensate for the

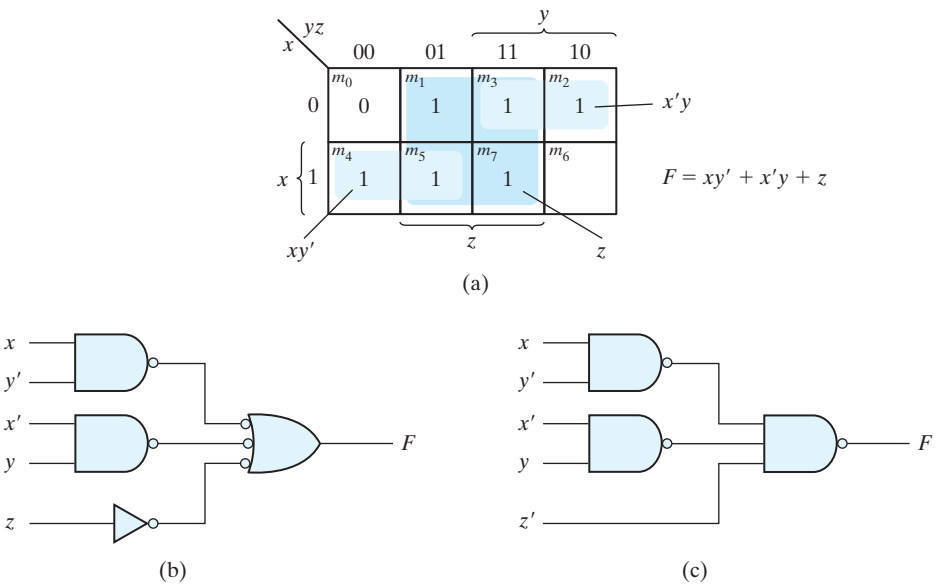


FIGURE 3.19
Solution to Example 3.9

bubble in the second-level gate. An alternative way of drawing the logic diagram is given in Fig. 3.19(c). Here, all the NAND gates are drawn with the same graphic symbol. The inverter with input z has been removed, but the input variable is complemented and denoted by z'.

■

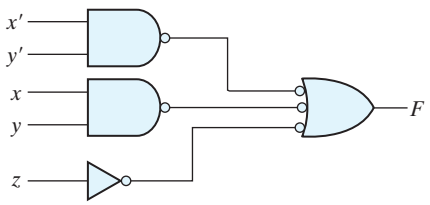
The procedure described in the previous example indicates that a Boolean function can be implemented with two levels of NAND gates. The procedure for obtaining the logic diagram from a Boolean function is as follows:

1. Simplify the function and express it in sum-of-products form.
2. Draw a NAND gate for each product term of the expression that has at least two literals. The inputs to each NAND gate are the literals of the term. This procedure produces a group of first-level gates.
3. Draw a single gate using the AND-invert or the invert-OR graphic symbol in the second level, with inputs coming from outputs of first-level gates.
4. A term with a single literal requires an inverter in the first level. However, if the single literal is complemented, it can be connected directly to an input of the second-level NAND gate.

Practice Exercise 3.10

Implement the Boolean function $F(x, y, z) = \Sigma(0, 1, 3, 5, 6, 7)$ with NAND gates, and draw the logic diagram of the implementation.

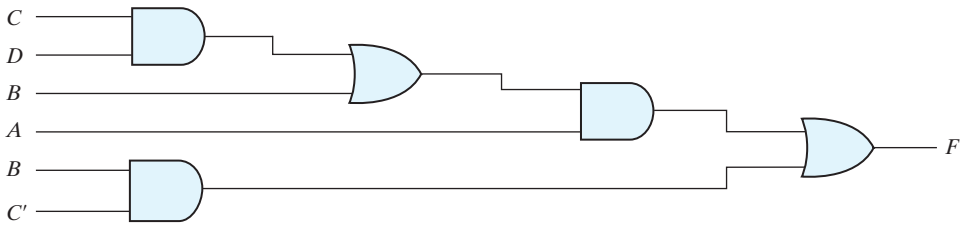
Answer: $F(x, y, z) = x'y' + xy + z$



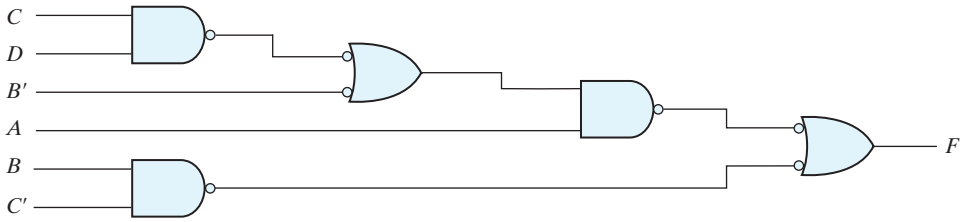
Multilevel NAND Circuits

The standard form of expressing Boolean functions results in a two-level implementation. There are occasions, however, when the design of digital systems results in gating structures with three or more levels. The most common procedure in the design of multilevel circuits is to express the Boolean function in terms of AND, OR, and complement operations. The function can then be implemented with AND and OR gates. After that, if necessary, it can be converted into an all-NAND circuit. Consider, for example, the Boolean function

$$F = A(CD + B) + BC'$$



(a) AND–OR gates



(b) NAND gates

FIGURE 3.20
Implementing $F = A(CD + B) + BC'$

Although it is possible to remove the parentheses and reduce the expression into a standard sum-of-products form, we choose to implement it as a multilevel circuit for illustration. The AND–OR implementation is shown in Fig. 3.20(a). There are four levels of gating in the circuit. The first level has two AND gates. The second level has an OR gate followed by an AND gate in the third level and an OR gate in the fourth level. A logic diagram with a pattern of alternating levels of AND and OR gates can easily be converted into a NAND circuit with the use of mixed notation, shown in Fig. 3.20(b). The procedure is to change every AND gate to an AND-invert graphic symbol and every OR gate to an invert-OR graphic symbol. The NAND circuit performs the same logic as the AND–OR diagram as long as there are two bubbles along the same line. The bubble associated with input B causes an extra complementation, which must be compensated for by changing the input literal to B'.

The general procedure for converting a multilevel AND–OR diagram into an all-NAND diagram using mixed notation is as follows:

1. Convert all AND gates to NAND gates with AND-invert graphic symbols.
2. Convert all OR gates to NAND gates with invert-OR graphic symbols.
3. Check all the bubbles in the diagram. For every bubble that is not compensated by another small circle along the same line, insert an inverter (a one-input NAND gate) or complement the input literal.

As another example, consider the multilevel Boolean function

$$F = (AB' + A'B)(C + D')$$

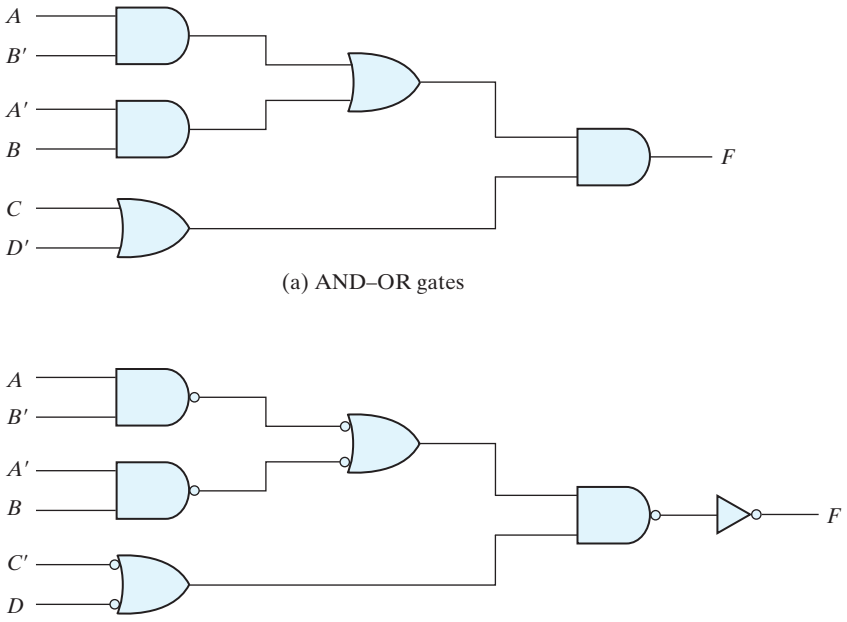


FIGURE 3.21
Implementing $F = (AB' + A'B)(C + D')$

The AND–OR implementation of this function is shown in Fig. 3.21(a) with three levels of gating. The conversion to NAND with mixed notation is presented in Fig. 3.21(b) of the diagram. The two additional bubbles associated with inputs C and D' cause these two literals to be complemented to C' and D. The bubble in the output NAND gate complements the output value, so we need to insert an inverter gate at the output in order to complement the signal again and get the original value back.

NOR Implementation

The NOR operation is the dual of the NAND operation. Therefore, all procedures and rules for NOR logic are the duals of the corresponding procedures and rules developed for NAND logic. The NOR gate is another universal gate that can be used to implement any Boolean function. The implementation of the complement, OR, and AND operations with NOR gates is shown in Fig. 3.22. The complement operation is obtained from a one-input NOR gate that behaves exactly like an inverter. The OR operation requires two NOR gates, and the AND operation is obtained with a NOR gate that has inverters in each input.

The two graphic symbols for the mixed notation are shown in Fig. 3.23. The OR-invert symbol defines the NOR operation as an OR followed by a complement. The invert-AND symbol complements each input and then performs an AND operation. The two symbols designate the same NOR operation and are logically identical because of DeMorgan's theorem.

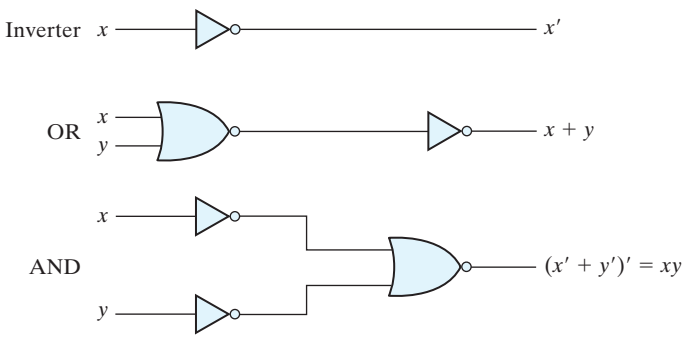
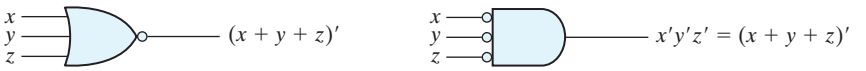


FIGURE 3.22
Logic operations with NOR gates



(a) OR-invert (b) Invert-AND

FIGURE 3.23
Two graphic symbols for the NOR gate

A two-level implementation with NOR gates requires that the function be simplified into product-of-sums form. Remember that the simplified product-of-sums expression is obtained from the map by combining the 0's and complementing. A product-of-sums expression is implemented with a first level of OR gates that produce the sum terms followed by a second-level AND gate to produce the product. The transformation from the OR–AND diagram to a NOR diagram is achieved by changing the OR gates to NOR gates with OR-invert graphic symbols and the AND gate to a NOR gate with an invert-AND graphic symbol. A single literal term going into the second-level gate must be complemented. Figure 3.24 shows the NOR implementation of a function expressed as a product of sums:

$$F = (A + B)(C + D)E$$

The OR–AND pattern can easily be detected by the removal of the bubbles along the same line. Variable E is complemented to compensate for the third bubble at the input of the second-level gate.

The procedure for converting a multilevel AND–OR diagram to an all-NOR diagram is similar to the one presented for NAND gates. For the NOR case, we must convert each OR gate to an OR-invert symbol and each AND gate to an invert-AND symbol. Any bubble that is not compensated by another bubble along the same line needs an inverter, or the complementation of the input literal.

The transformation of the AND–OR diagram of Fig. 3.21(a) into a NOR diagram is shown in Fig. 3.25. The Boolean function for this circuit is

$$F = (AB' + A'B)(C + D')$$

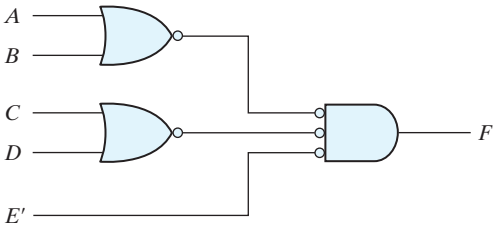


FIGURE 3.24
Implementing $F = (A + B)(C + D)E$

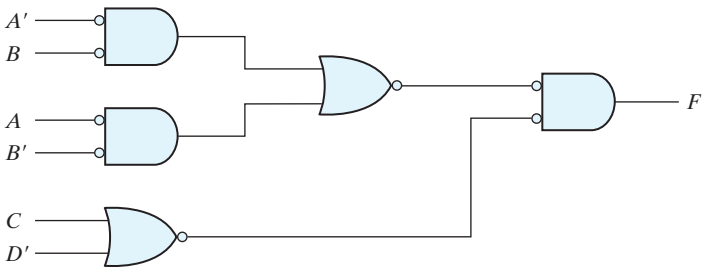


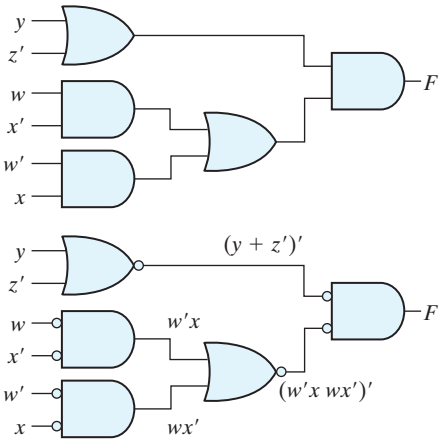
FIGURE 3.25
Implementing $F = (AB' + A'B)(C + D')$ with NOR gates

The equivalent AND–OR diagram can be recognized from the NOR diagram by removing all the bubbles. To compensate for the bubbles in four inputs, it is necessary to complement the corresponding input literals.

Practice Exercise 3.11

Implement the Boolean function $F(w, x, y, z) = (y + z')(wx' + w'x)$ with NOR gates.

Answer:



3.7 OTHER TWO-LEVEL IMPLEMENTATIONS

The types of gates most often found in integrated circuits are NAND and NOR gates. For this reason, NAND and NOR logic implementations are the most important from a practical point of view. Some (but not all) NAND or NOR gates allow the possibility of a wire connection between the outputs of two gates to provide a specific logic function. This type of logic is called *wired logic*. For example, open-collector TTL NAND gates, when tied together, perform wired-AND logic. The wired-AND logic performed with two NAND gates is depicted in Fig. 3.26(a). The AND gate is drawn with the lines going through the center of the gate to distinguish it from a conventional gate. The wired-AND gate is not a physical gate, but only a symbol to designate the function obtained from the indicated wired connection. The logic function implemented by the circuit of Fig. 3.26(a) is

$$F = (AB)'(CD)' = (AB + CD)' = (A' + B')(C' + D')$$

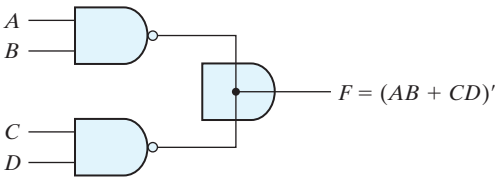
and is called an AND–OR–INVERT function.

Similarly, the NOR outputs of ECL gates can be tied together to perform a wired-OR function. The logic function implemented by the circuit of Fig. 3.26(b) is

$$F = (A + B)' + (C + D)' = [(A + B)(C + D)]',$$

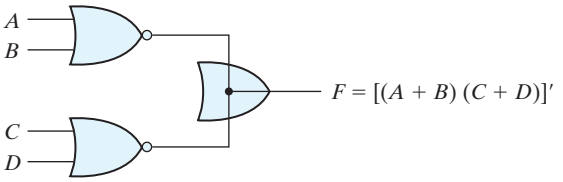
and is called an OR–AND–INVERT function.

A wired-logic gate does not produce a physical second-level gate, since it is just a wire connection. Nevertheless, for discussion purposes, we will consider the circuits of Fig. 3.26 as two-level implementations. The first level consists of NAND (or NOR) gates and the second level has a single AND (or OR) gate. The wired connection in the graphic symbol will be omitted in subsequent discussions.[3]



(a) Wired-AND in open-collector TTL NAND gates.

(AND–OR–INVERT)



(b) Wired-OR in ECL gates

(OR–AND–INVERT)

FIGURE 3.26
Wired logic
(a) Wired-AND logic with two NAND gates
(b) Wired-OR in emitter-coupled logic (ECL) gates

[3]The family of nets in the Verilog HDL includes two wired net types: **wand** and **wor**. A **wand** net is driven to logical 0 if any of its drivers is 0; a **wor** net is driven to 1 if any of its drivers is 1. We will not make use of these nets.

Nondegenerate Forms

It will be instructive from a theoretical point of view to find out how many two-level combinations of gates are possible. We consider four types of gates: AND, OR, NAND, and NOR. If we assign one type of gate for the first level and one type for the second level, we find that there are 16 possible combinations of two-level forms. (The same type of gate can be in the first and second levels, as in a NAND–NAND implementation.) Eight of these combinations are said to be *degenerate* forms because they degenerate to a single operation. This can be seen from a circuit with AND gates in the first level and an AND gate in the second level. The output of the circuit is merely the AND function of all input variables. The remaining eight *nondegenerate* forms produce an implementation in sum-of-products form or product-of-sums form. The eight *nondegenerate* forms are as follows:

AND–OR	OR–AND
NAND–NAND	NOR–NOR
NOR–OR	NAND–AND
OR–NAND	AND–NOR

The first gate listed in each of the forms constitutes a first level in the implementation. The second gate listed is a single gate placed in the second level. Note that any two forms listed on the same line are duals of each other.

The AND–OR and OR–AND forms are the basic two-level forms discussed in Section 3.4. The NAND–NAND and NOR–NOR forms were presented in Section 3.6. The remaining four forms are investigated in this section.

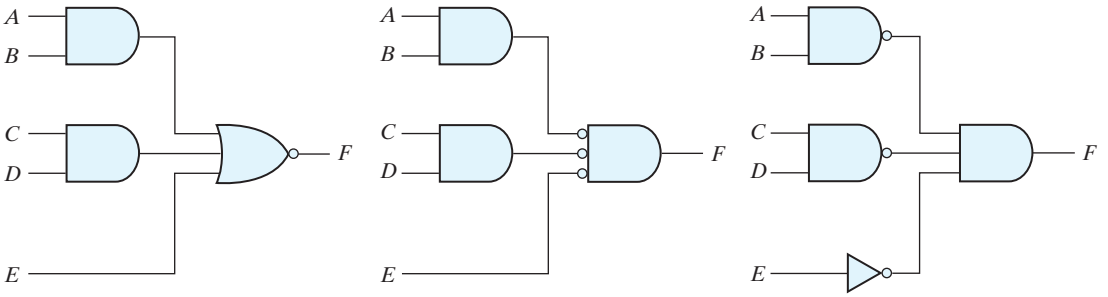
AND–OR–INVERT Implementation

The two forms, NAND–AND and AND–NOR, are equivalent and can be treated together. Both perform the AND–OR–INVERT function, as shown in Fig. 3.27. The AND–NOR form resembles the AND–OR form, but with an inversion done by the bubble in the output of the NOR gate. It implements the function

$$F = (AB + CD + E)'$$

By using the alternative graphic symbol for the NOR gate, we obtain the diagram of Fig. 3.27(b). Note that the single variable E is *not* complemented, because the only change made is in the graphic symbol of the NOR gate. Now we move the bubble from the input terminal of the second-level gate to the output terminals of the first-level gates. An inverter is needed for the single variable in order to compensate for the bubble. Alternatively, the inverter can be removed, provided that input E is complemented. The circuit of Fig. 3.27(c) is a NAND–AND form and was shown in Fig. 3.26 to implement the AND–OR–INVERT function.

An AND–OR implementation requires an expression in sum-of-products form. The AND–OR–INVERT implementation is similar, except for the inversion. Therefore, if the *complement* of the function is simplified into sum-of-products form (by combining



(a) AND–NOR (b) AND–NOR (c) NAND–AND

FIGURE 3.27
AND–OR–INVERT circuits, $F = (AB + CD + E)'$

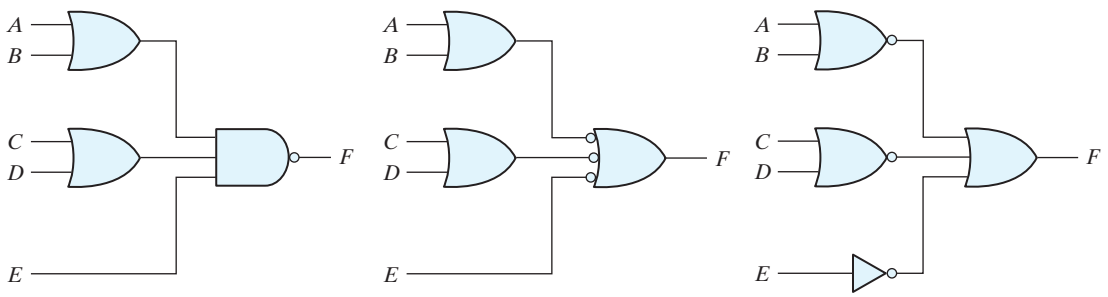
the 0's in the map), it will be possible to implement F' with the AND–OR part of the function. When F' passes through the always present output inversion (the INVERT part), it will generate the output F of the function. An example for the AND–OR–INVERT implementation will be shown subsequently.

OR–AND–INVERT Implementation

The OR–NAND and NOR–OR forms perform the OR–AND–INVERT function, as shown in Fig. 3.28. The OR–NAND form resembles the OR–AND form, except for the inversion done by the bubble in the NAND gate. It implements the function

$$F = [(A + B)(C + D)E]'$$

By using the alternative graphic symbol for the NAND gate, we obtain the diagram of Fig. 3.28(b). The circuit in Fig. 3.28(c) is obtained by moving the small circles from the inputs of the second-level gate to the outputs of the first-level gates. The circuit of



(a) OR–NAND (b) OR–NAND (c) NOR–OR

FIGURE 3.28
OR–AND–INVERT circuits, $F = [(A + B)(C + D)E]'$

Table 3.2
Implementation with Other Two-Level Forms

Equivalent Nondegenerate Implementation		Implements the Form	Simplify F' into	To Get an Output of
(a)	(b)*			
AND–NOR	NAND–AND	AND–OR–INVERT	Sum-of-products form by combining 0's in the map.	F
OR–NAND	NOR–OR	OR–AND–INVERT	Product-of-sums form by combining 1's in the map and then complementing.	F

*Form (b) requires an inverter for a single literal term.

Fig. 3.28(c) is a NOR–OR form and was shown in Fig. 3.26 to implement the OR–AND–INVERT function.

The OR–AND–INVERT implementation requires an expression in product-of-sums form. If the complement of the function is simplified into that form, we can implement F' with the OR–AND part of the function. When F' passes through the INVERT part, we obtain the complement of F', or F, in the output.

Tabular Summary and Example

Table 3.2 summarizes the procedures for implementing a Boolean function in any one of the four 2-level forms. Because of the INVERT part in each case, it is convenient to use the simplification of F' (the complement) of the function. When F' is implemented in one of these forms, we obtain the complement of the function in the AND–OR or OR–AND form. The four 2-level forms invert this function, giving an output that is the complement of F'. This is the normal output F.

EXAMPLE 3.10

Implement the function of Fig. 3.29(a) with the four 2-level forms listed in Table 3.2.

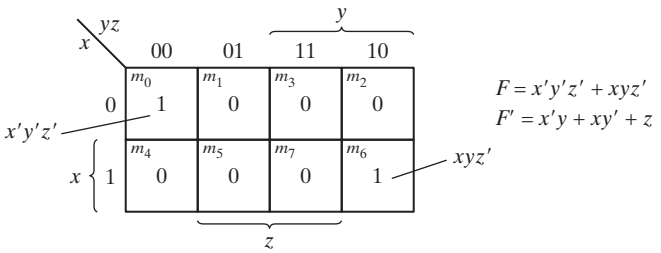
The complement of the function is simplified into sum-of-products form by combining the 0's in the map:

$$F' = x'y + xy' + z$$

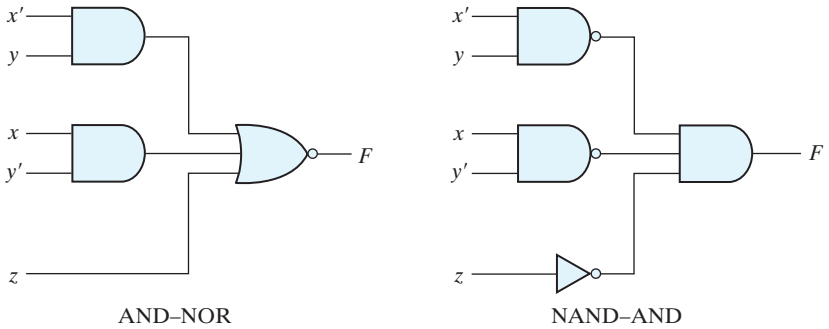
The normal output for this function can be expressed as

$$F = (x'y + xy' + z)'$$

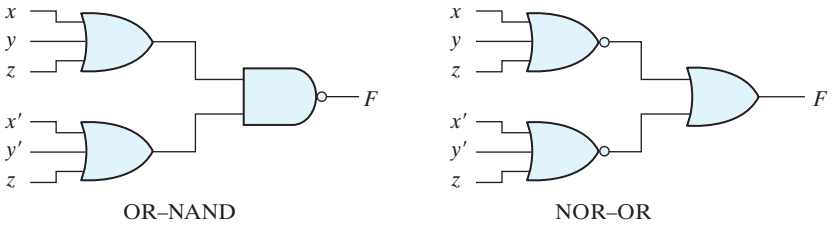
which is in the AND–OR–INVERT form. The AND–NOR and NAND–AND implementations are shown in Fig. 3.29(b). Note that a one-input NAND, or inverter, gate is needed in the NAND–AND implementation, but not in the AND–NOR case. The inverter can be removed if we apply the input variable z' instead of z.



(a) Map simplification in sum of products



(b) $F = (x'y + xy' + z)'$



(c) $F = [(x + y + z)\,(x' + y' + z)]'$

FIGURE 3.29
Other two-level implementations

The OR–AND–INVERT forms require a simplified expression of the complement of the function in product-of-sums form. To obtain this expression, we first combine the 1's in the map:

$$F = x'y'z' + xyz'$$

Then we take the complement of the function:

$$F' = (x + y + z)(x' + y' + z)$$

The normal output F can now be expressed in the form

$$F = [(x + y + z)(x' + y' + z)]'$$

which is the OR–AND–INVERT form. From this expression, we can implement the function in the OR–NAND and NOR–OR forms, as shown in Fig. 3.29(c).

■

3.8 EXCLUSIVE-OR FUNCTION

The exclusive-OR (XOR), denoted by the symbol $\oplus$, is a logical operation that performs the following Boolean operation:

$$x \oplus y = xy' + x'y$$

The exclusive-OR is equal to 1 if only x is equal to 1 or if only y is equal to 1 (i.e., x and y differ in value), but not when both are equal to 1 or when both are equal to 0. The exclusive-NOR, also known as equivalence, performs the following Boolean operation:

$$(x \oplus y)' = xy + x'y'$$

The exclusive-NOR is equal to 1 if both x and y are equal to 1 or if both are equal to 0. The exclusive-NOR can be shown to be the complement of the exclusive-OR by means of a truth table or by algebraic manipulation:

$$(x \oplus y)' = (xy' + x'y)' = (x' + y)(x + y') = xy + x'y'$$

The following identities apply to the exclusive-OR operation:

$$x \oplus 0 = x$$
$$x \oplus 1 = x'$$
$$x \oplus x = 0$$
$$x \oplus x' = 1$$
$$x \oplus y' = x' \oplus y = (x \oplus y)'$$

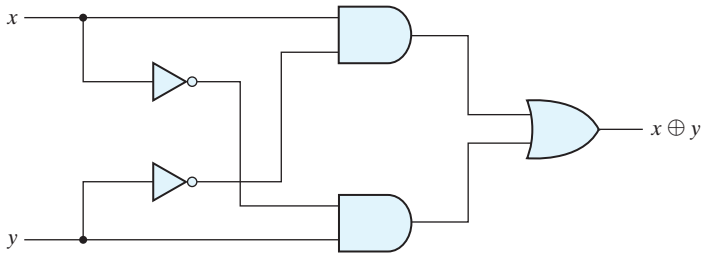
Any of these identities can be proven with a truth table or by replacing the $\oplus$ operation by its equivalent Boolean expression. Also, it can be shown that the exclusive-OR operation is both commutative and associative; that is,

$$A \oplus B = B \oplus A$$

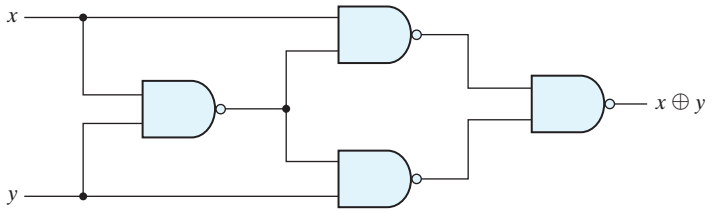
and

$$(A \oplus B) \oplus C = A \oplus (B \oplus C) = A \oplus B \oplus C$$

This means that the two inputs to an exclusive-OR gate can be interchanged without affecting the operation. It also means that we can evaluate a three-variable exclusive-OR operation in any order, and for this reason, three or more variables can be expressed without parentheses. This would imply the possibility of using exclusive-OR gates with



(a) Exclusive-OR with AND–OR–NOT gates



(b) Exclusive-OR with NAND gates

FIGURE 3.30
Logic diagrams for exclusive-OR implementations

three or more inputs. However, multiple-input exclusive-OR gates are difficult to fabricate with hardware. In fact, even a two-input function is usually constructed with other types of gates. A two-input exclusive-OR function is constructed with conventional gates using two inverters, two AND gates, and an OR gate, as shown in Fig. 3.30(a). Figure 3.30(b) shows the implementation of the exclusive-OR with four NAND gates. The first NAND gate performs the operation $(xy)' = (x' + y')$. The other two-level NAND circuit produces the sum of products of its inputs:

$$(x' + y')\,x + (x' + y')\,y = xy' + x'y = x \oplus y$$

Only a limited number of Boolean functions can be expressed in terms of exclusive-OR operations. Nevertheless, this function emerges quite often during the design of digital systems. It is particularly useful in arithmetic operations and error detection and correction circuits.

Odd Function

The exclusive-OR operation with three or more variables can be converted into an ordinary Boolean function by replacing the $\oplus$ symbol with its equivalent Boolean expression. In particular, the three-variable case can be converted to a Boolean expression as follows:

$$\begin{aligned} A \oplus B \oplus C &= (AB' + A'B)C' + (AB + A'B')C \\ &= AB'C' + A'BC' + ABC + A'B'C \\ &= \Sigma(1, 2, 4, 7) \end{aligned}$$

The Boolean expression clearly indicates that the three-variable exclusive-OR function is equal to 1 if only one variable is equal to 1 or if all three variables are equal to 1. Contrary to the two-variable case, in which only one variable must be equal to 1, in the case of three or more variables the requirement is that an odd number of variables be equal to 1. As a consequence, the multiple-variable exclusive-OR operation is defined as an *odd function.*

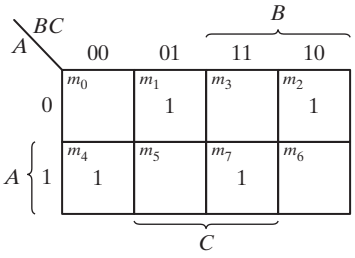
The Boolean function derived from the three-variable exclusive-OR operation is expressed as the logical sum of four minterms whose binary numerical values are 001, 010, 100, and 111. Each of these binary numbers has an odd number of 1's. The remaining four minterms not included in the function are 000, 011, 101, and 110, and they have an even number of 1's in their binary numerical values. In general, an n-variable exclusive-OR function is an odd function defined as the logical sum of the $2^n/2$ minterms whose binary numerical values have an odd number of 1's.

The definition of an odd function can be clarified by plotting it in a map. Figure 3.31(a) shows the map for the three-variable exclusive-OR function. The four minterms of the function are a unit distance apart from each other. The odd function is identified from the four minterms whose binary values have an odd number of 1's. The complement of an odd function is an even function. As shown in Fig. 3.31(b), the three-variable even function is equal to 1 when an even number of its variables is equal to 1 (including the condition that none of the variables is equal to 1).

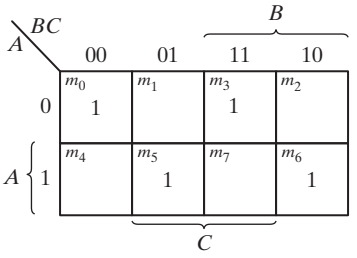
The three-input odd function is implemented by means of two-input exclusive-OR gates, as shown in Fig. 3.32(a). The complement of an odd function is obtained by replacing the output gate with an exclusive-NOR gate, as shown in Fig. 3.32(b).

Consider now the four-variable exclusive-OR operation. By algebraic manipulation, we can obtain the sum of minterms for this function:

$$\begin{aligned} A \oplus B \oplus C \oplus D &= (AB' + A'B) \oplus (CD' + C'D) \\ &= (AB' + A'B)(CD + C'D') + (AB + A'B')(CD' + C'D) \\ &= \Sigma(1, 2, 4, 7, 8, 11, 13, 14) \end{aligned}$$

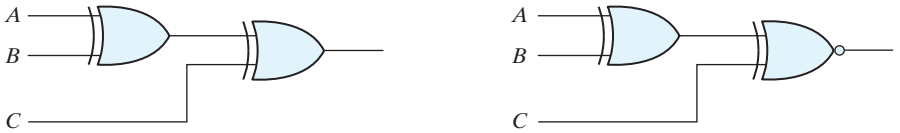


(a) Odd function $F = A \oplus B \oplus C$



(b) Even function $F = (A \oplus B \oplus C)'$

FIGURE 3.31
Map for a three-variable exclusive-OR function



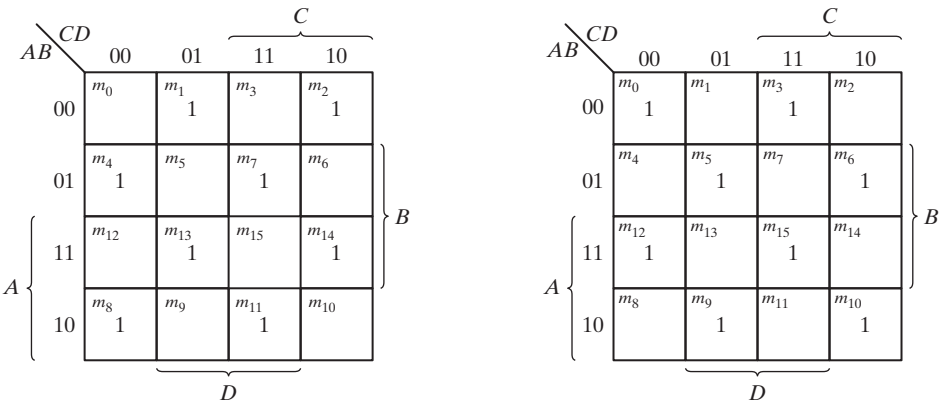
(a) 3-input odd function (b) 3-input even function

FIGURE 3.32
Logic diagram of odd and even functions

There are 16 minterms for a four-variable Boolean function. Half of the minterms have binary numerical values with an odd number of 1's; the other half of the minterms have binary numerical values with an even number of 1's. In plotting the function in the map, the binary numerical value for a minterm is determined from the row and column numbers of the square that represents the minterm. The map of Fig. 3.33(a) is a plot of the four-variable exclusive-OR function. This is an odd function because the binary values of all the minterms have an odd number of 1's. The complement of an odd function is an even function. As shown in Fig. 3.33(b), the four-variable even function is equal to 1 when an even number of its variables is equal to 1.

Parity Generation and Checking

Exclusive-OR functions are very useful in systems requiring error detection and correction codes. As discussed in Section 1.7, a parity bit is used for the purpose of detecting errors during the transmission of binary information. A parity bit is an extra bit included with a binary message to make the number of 1's either odd or even. The message, including the parity bit, is transmitted and then checked at the receiving end for errors.



(a) Odd function $F = A \oplus B \oplus C \oplus D$ (b) Even function $F = (A \oplus B \oplus C \oplus D)'$

FIGURE 3.33
Map for a four-variable exclusive-OR function

Table 3.3
Even-Parity-Generator Truth Table

Three-Bit Message			Parity Bit
x	*y*	*z*	*P*
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

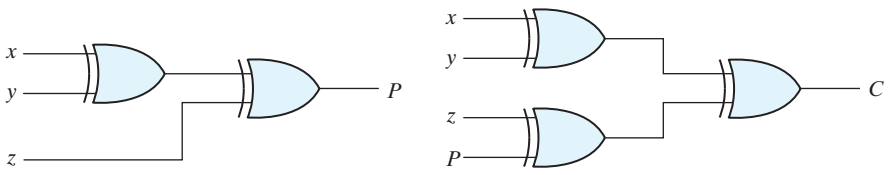
An error is detected if the checked parity does not correspond with the one transmitted. The circuit that generates the parity bit in the transmitter is called a *parity generator.* The circuit that checks the parity in the receiver is called a *parity checker.*

As an example, consider a three-bit message to be transmitted together with an even-parity bit. Table 3.3 shows the truth table for the parity generator. The three bits—x, y, and z—constitute the message and are the inputs to the circuit. The parity bit P is the output. For even-parity, the bit P must be generated to make the total number of 1's (including P) even. From the truth table, we see that P constitutes an odd function because it is equal to 1 for those minterms whose numerical values have an odd number of 1's. Therefore, P can be expressed as a three-variable exclusive-OR function:

$$P = x \oplus y \oplus z$$

The logic diagram for the parity generator is shown in Fig. 3.34(a).

The three bits in the message, together with the parity bit, are transmitted to their destination, where they are applied to a parity-checker circuit to check for possible errors in the transmission. Since the information was transmitted with even parity, the four bits received must have an even number of 1's. An error occurs during the transmission if the four bits received have an odd number of 1's, indicating that one bit has changed in value during transmission. The output of the parity checker, denoted by C, will be



(a) 3-bit even-parity generator (b) 4-bit even-parity checker

FIGURE 3.34
Logic diagram of a parity generator and checker

equal to 1 if an error occurs—that is, if the four bits received have an odd number of 1's. Table 3.4 is the truth table for the even-parity checker. From it, we see that the function C consists of the eight minterms with binary numerical values having an odd number of 1's. The table corresponds to the map of Fig. 3.33(a), which represents an odd function. The parity checker can be implemented with exclusive-OR gates:

$$C = x \oplus y \oplus z \oplus P$$

The logic diagram of the parity checker is shown in Fig. 3.34(b).

It is worth noting that the parity generator can be implemented with the circuit of Fig. 3.34(b) if the input P is connected to logic 0 and the output is marked with P. This is because $z \oplus 0 = z$, causing the value of z to pass through the gate unchanged. The advantage of this strategy is that the same circuit can be used for both parity generation and checking.

It is obvious from the foregoing example that parity generation and checking circuits always have an output function that includes half of the minterms whose numerical values have either an odd or even number of 1's. As a consequence, they can be implemented with exclusive-OR gates. A function with an even number of 1's is the complement of an odd function. It is implemented with exclusive-OR gates, except that the gate associated with the output must be an exclusive-NOR to provide the required complementation.

Table 3.4
Even-Parity-Checker Truth Table

Four Bits Received				Parity Error Check
x	*y*	*z*	*P*	C
0	0	0	0	0
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	1
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0

3.9 HARDWARE DESCRIPTION LANGUAGES (HDLs)

Manual methods for designing logic circuits are feasible only when the circuit is small. For anything else (i.e., a practical circuit), designers use computer-based design tools to reduce costs and minimize the risk of creating a flawed design. Prototype integrated circuits are too expensive and time consuming to build, so all modern design tools rely on a hardware description language to describe, design, and test a circuit in software before it is ever manufactured.

A *hardware description language* (HDL) is a computer-based language that describes the hardware of digital systems in a textual form. Before the advent of HDLs, designers relied on schematics of block diagrams and logic gates to represent and specify a circuit. That methodology is prone to error and its results are costly to edit, especially for complex circuits. In contrast, today's HDL-based design tools create an HDL description, then derive a schematic automatically and correctly, as a by-product of the design methodology. Revisions of the HDL description simplify the creation and revision of a schematic.

An HDL is a *modeling* language rather than a *computational* language. An HDL resembles an ordinary computer programming language, such as C, but is specifically oriented to describing hardware structures and the behavior of logic circuits. It can be used to represent logic diagrams, truth tables, Boolean expressions, and complex abstractions of the behavior of a digital system. Those features distinguish an HDL from other types of languages, many of which are used to perform computations on numerical data. One way to view an HDL is to observe that it *describes a relationship between signals that are the inputs to a circuit and the signals that are the outputs of the circuit*. For example, an HDL description of an AND gate describes how the logic value of the gate's output is determined by the logic values of its inputs.

As a *documentation* language, an HDL is used to represent and document digital systems in a form that can be read by both humans and computers and is suitable as an exchange language between designers. The language content can be stored, retrieved, edited, and transmitted easily and processed by computer software in an efficient manner.

HDLs are used in several major steps in the design flow of an integrated circuit: design entry, functional simulation or verification, logic synthesis, timing verification, and fault simulation.

Design entry creates an HDL-based description of the functionality that is to be implemented in hardware. Depending on the HDL, the description can be in a variety of forms: Boolean logic equations, truth tables, a net list of interconnected gates, or an abstract behavioral model. The HDL model may also represent a partition of a larger circuit into smaller interconnected and interacting functional units.

Logic simulation displays the behavior of a digital system through the use of a computer. A simulator interprets the HDL description and either produces readable output, such as a time-ordered sequence of input and output signal values, or displays waveforms of the signals. The simulation of a circuit shows how the hardware will behave before it is actually fabricated. Simulation detects functional errors in a design without having to physically create and operate the circuit. Errors that are detected during a simulation can be corrected by modifying the appropriate HDL statements. The stimulus

(i.e., the logic values of the inputs to a circuit) that tests the functionality of the design is called a *test bench*. Thus, to simulate a digital system, the design is first described in an HDL and then verified by simulating the design and checking it with a *test bench*, which is also written in the HDL. An alternative and more complex approach relies on formal mathematical methods to prove that a circuit is functionally correct. That approach is beyond the level of this text. We will focus exclusively on simulation.

Logic synthesis derives an optimized list of physical components and their interconnections (called a *netlist*) from the model of a digital system described in an HDL. The netlist can be used to fabricate an integrated circuit or to lay out a printed circuit board with the hardware counterparts of the gates in the list. Logic synthesis produces a database describing the elements and structure of a circuit. It specifies how to fabricate a physical integrated circuit that implements in silicon the functionality described by statements made in an HDL. Logic synthesis (1) is based on formal procedures that implement digital circuits, and (2) performs logic minimization on those parts of a digital design process, which can be automated with computer software. The design of today's large, complex circuits is made possible by logic synthesis software. It is essential that users of an HDL realize that not all constructs of the language are synthesizable.

Timing verification confirms that a synthesized and fabricated, integrated circuit will operate at a specified speed. Because each logic gate in a circuit has a propagation delay, a signal transition at the input of a circuit cannot immediately cause a change in the logic value of the output of a circuit. Propagation delays ultimately limit the speed at which a circuit can operate. Timing verification checks each signal path to verify that it is not compromised by propagation delay. This step is done after logic synthesis specifies the actual devices that will compose a circuit and before the implementation is released for production.

In VLSI circuit design, *fault simulation* compares the behavior of an ideal circuit with the behavior of a circuit that contains a process-induced flaw. Dust and other particulates in the atmosphere of the clean room can cause a circuit to be fabricated with a fault. A circuit with a fault will not exhibit the same functionality as a fault-free circuit. Fault simulation is used to identify input stimuli that can be used to reveal the difference between the faulty circuit and the fault-free circuit. These test patterns will be used to test fabricated devices to ensure that only good devices are shipped to the customer. Test generation and fault simulation may occur at different steps in the design process, but they are always done before production in order to avoid the disaster of producing a circuit whose internal logic cannot be tested.

Design Encapsulation and Modeling with HDLs

Companies that design integrated circuits use proprietary and public HDLs. In the public domain, the IEEE supports the following standardized HDLs: VHDL, Verilog, and System Verilog. VHDL is a U.S. Department of Defense-mandated language.[4] The path

[4]The V in VHDL stands for the first letter in VHSIC, an acronym for Very High Speed Integrated Circuit.

of development of Verilog ultimately led to its being a proprietary HDL of Cadence Design Systems, which transferred control of Verilog to a consortium of companies and universities known as Open Verilog International (OVI)[5] as a step leading to its adoption as an IEEE standard. SystemVerilog evolved mainly from Verilog and from Super Log, a proprietary language held by Synopsys, Inc. The Verilog-2005 language is embedded within SystemVerilog, so *our text will consider Verilog before presenting a brief introduction to SystemVerilog.*

Design encapsulation, or *design entry*, creates a model representing the functionality of a digital circuit. The model is a repository for the features that determine the behavior of a circuit and, possibly, its structure. The reference manual of each language governs how models may be constructed.

Verilog—Design Encapsulation

The language reference manual for the Verilog HDL presents the syntax that describes precisely the constructs of the language. A Verilog model is composed of text using keywords, of which there are about 100. Keywords are predefined lowercase identifiers that define the language constructs. Examples of keywords are **module**, **endmodule**, **input**, **output**, **wire**, **and**, **or**, and **not**. For emphasis and clarity, keywords will be identified in the text by displaying them in boldface in all examples of code and wherever it is helpful to call attention to their use. Lines of text terminate with a semicolon (;), and any text between two forward slashes (//) and the end of the line is interpreted as a comment. A comment has no effect on a simulation using the model. Multiline comments begin with /* and terminate with */. They may not be nested. Blank spaces are ignored, but they may not appear within the text of a keyword, a user-specified identifier, an operator, or the representation of a number. *Verilog is case sensitive*, which means that uppercase and lowercase letters are distinguishable (e.g., **not** is not the same as NOT).

The term *module* refers to the text enclosed by the keyword pair **module** . . . **endmodule**. A module is the fundamental descriptive (design) unit in the Verilog language. It is declared by the keyword **module** and must always be terminated by the keyword **endmodule**.

Previous sections of the text have demonstrated that combinational logic can be described by a set of Boolean equations, by a schematic connection of gates, or by a truth table. Now we'll consider how HDLs implement these descriptions of combinational logic.

Verilog Example 3.1

Figure 3.35 is a logic diagram for a simple circuit in which the output of an OR gate is one of two inputs to an AND gate. The Boolean equation for the output of the circuit

[5] OVI evolved to become Accellera—see www.accellera.org.

can be written directly from the diagram: E = (A + B)C. We'll use its Verilog description to introduce key details of the language.

```
module or_and (
 output      E,
 input       A, B, C
);
 wire D;
 assign D = A || B;        // | is logical "OR" operator
 assign E = C && D;        // & is the logical "AND" operator
// This is a single-line comment
/* The text here and below
   form a multi-line comment
*/
endmodule
```

The Verilog description of the circuit begins with the keyword **module** and the name of the design (*or_and*).[6] The keyword **module** starts the declaration of the description; the last line completes the declaration with the keyword **endmodule**. The keyword **module** is followed by the name and a parenthesis-enclosed list of ports.

The name of a design unit is an *identifier*. Identifiers are names given to modules, variables (e.g., a signal), and other elements of the language so that they can be distinguished and referenced in the design. In general, we choose meaningful names for modules. Identifiers are composed of alphanumeric characters and the underscore (_), and are case sensitive. Identifiers must start with an alphabetic character or an underscore, but they may not start with a number.

Boolean equations like those in the previous chapters describe the input–output logic of a digital circuit. In Verilog they are composed as *continuous assignment* statements and placed within the code space defined by the **module** . . . **endmodule** keywords.

A continuous assignment statement has the appearance of an equation, but it is essential to understand that *a continuous assignment does not prescribe a computation.* Instead, it defines a *relationship* between signals in a circuit. Consider the Boolean equations $D = A + B$ and $E = CD$, corresponding to the schematic in Fig. 3.35, where A, B, C, D, and E are Boolean variables.

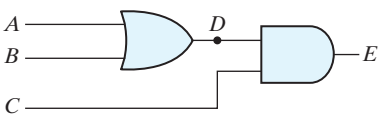


FIGURE 3.35
A logic diagram (schematic) for the Boolean equations $D = A + B$ $E = CD$

[6] It is a common practice to place each module in a file having the same name as the module. The filename extension would be .v.

In the Verilog code, signal *D* is formed by the "OR" of inputs *A* and *B*; output *E* is formed as the "AND" of *C* and *D*. The continuous assignment is specified by the keyword **assign,** followed by a Boolean expression; the assignment is *continuous* in the sense that it always (i.e., for the duration of a simulation) governs the relationship between *D* and *A* and *B*, and between *E* and *C* and *D*, just as the output of a logic gate is always determined by the inputs to the gate and the function of the gate. Verilog uses the logic operator symbols &&, ||, and ! to represent the logical operators AND, OR, and NOT, respectively. These keywords are not logic gates, but a synthesis tool may associate gates with them.

The *port list* of a module is the interface between the module and its environment. In the model *or_and*, the ports are the inputs (A, B, C) and the output (E) of the circuit. The *mode*, or direction, of a port distinguishes between *inputs*, *outputs*, and *inouts* (bidirectional) ports. The logic values of the inputs to a circuit are determined by the environment; the logic values of the outputs are determined within the circuit and result from the action of the inputs on the circuit. The logic value of an inout port may be determined by the environment or by the internal logic of the module. The port list is enclosed in parentheses, and commas are used to separate elements of the list. The statement is terminated with a semicolon (;). In our examples, all keywords (which must be in lowercase) are printed in bold for clarity, but that is not a requirement of the language. Next, the keywords **input** and **output** specify which of the ports are inputs and which are outputs. Internal connections, such as D, are declared as wires. Note that, with correct interpretation of the language operators, the continuous assignment statements in the model implicitly describe the schematic shown in Fig. 3.35.

Practice Exercise 3.12 – Verilog

Write a continuous assignment statement that describes *Y* in the logic diagram in Fig. PE3.12, where *A*, *B*, *C*, and *D* are Boolean variables.

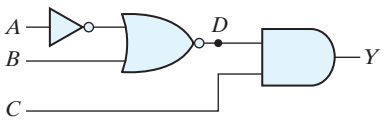


FIGURE PE3.12

Answer: assign Y = (!((!A)||B))&&C

Verilog Example 3.2

This example develops a Verilog model of a circuit having inputs *A, B, C, D* and outputs *E, F*, with functionality specified by the following Boolean expressions:

$$E = A + BC + B'D$$
$$F = B'C + BC'D'$$

Answer:

```
module Circuit_Boolean_CA (E, F, A, B, C, D);
 output      E, F;
 input       A, B, C, D;
 assign E = A || (B && C) || ((!B) && D);
 assign F = ((!B) && C) || (B && (!C) && (!D));
endmodule
```

Two continuous assignment statements describe the Boolean equations for E and F. This description illustrates that the declaration of the modes of the ports may follow the port list, rather than be included in it. The values of E and F during simulation are determined dynamically by the values of A, B, C, and D. A simulator will detect when the test bench changes a value of one or more of the inputs. When this happens, the simulator updates, if necessary, the values of E and F. The continuous assignment mechanism is so named because the relationship between the assigned value and the variables is permanent. The mechanism acts just like combinational logic, and has a gate-level equivalent circuit.

VHDL—Design Encapsulation

A design encapsulation in VHDL has two parts: an **entity** and an **architecture**. A VHDL **entity** (1) provides a name by which a design can be identified, and (2) specifies the interface of the design with its environment. The name and direction (i.e., *mode*) of each interface signal and its data type are declared in the port of the entity. The syntax template of an *entity* is given below:

```
entity name_of_entity is
 port (names_of_signals : mode_of_signals signal_type;
       names_of_signals : mode_of_signals signal_type;
                        . . .
       names_of_signals : mode_of_signals signal_type);
end name_of_entity;
```

A simple example is given by:

```
entity Simple_Example is
 port (y_out: out bit; x_in: in bit);
end Simple_Example;
```

Any architecture that is paired with the entity can use the declared port signals to describe the logic it represents, without having to re-declare the signals. An identifier that appears in a port is implicitly a signal.[7] Signals are implemented in hardware as the

[7] VHDL also has variables, like other software languages, but only signals may be declared in a port.

electrical connections of a circuit, and they represent the logical data that is processed by a circuit. The syntax template for a declaration of a *signal* is defined as

```
signal list_of_signal_names: signal_type;
```

The identifiers that are declared in the port of an entity are implicitly signals, and are available to any architecture associated with the entity. A signal declared within an architecture is local to the architecture in which it is declared, that is, it can be referenced only within that architecture. An output signal in the port of an entity can be read externally.

The functional description of a design is provided by an **architecture**, which describes how the outputs of an entity are formed from its inputs. A given design may have a variety of descriptions, allowing more than one architecture to be associated with an entity. An architecture has the following syntax template:

```
architecture architecture_name of  entity_name is
    declarations_of_data_types
    declarations_of_signals
    declarations_of_constants
    definitions_of_functions
    definitions_of_procedures
    declarations_of_components
begin
    concurrent_statements
 end [architecture][8] architecture_name;
```

The concurrent statements that may be declared within an architecture are (1) component instantiations, (2) signal assignments, and (3) process statements.

VHDL is not a case sensitive language. Language keywords are shown in bold font in the VHDL text only for emphasis.

VHDL Example 3.1

Figure 3.36 depicts *or_and_vhdl*, a VHDL entity-architecture pair for a simple logic circuit. The entity identifies and specifies the mode and type of all of the inputs and outputs of the circuit. In VHDL the keyword names of allowed port modes (directions) are **in**, **out**, **inout**, and **buffer**. An **inout** port is bidirectional—one whose value can be generated within the architecture of the module and externally as well. A **buffer** declares that the port is an output but is also read within the module, for example, in a signal assignment.

The entity *or_and_vhdl* provides an interface between the architecture and its external environment. In this example the interface is a **port** consisting of three named *input* signals (*A, B, and C*) and one *output* signal (*E*). In general, a **port** statement identifies the input and output signals of the circuit, their direction (**in**, **out, inout, buffer**), and their data type (e.g., std_logic). Signals may be listed in any order in a port. The architecture here includes an internal signal, *D*, having type std_logic. Signal *D* connects the output

[8] [architecture] is an optional item in the closing statement.

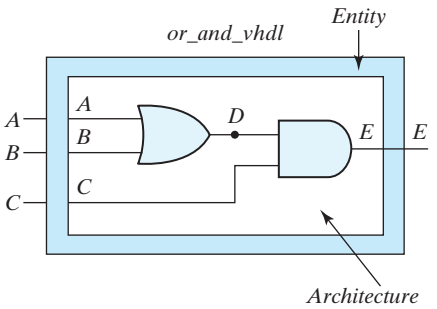


FIGURE 3.36
Entity-Architecture pair for *or_and_vhdl*

of the OR gate to an input of the AND gate, but is not part of the entity because it does not connect to the world outside *or_and_vhdl*. Signal *D* is not visible at the interface to the environment.

The descriptive style used in this architecture (below); is based on Boolean equations. It uses VHDL's built-in data operators[9] to declare signal assignments using the Boolean expressions and equations implied by the schematic in Fig. 3.36. The signal assignment operator (<=) and the accompanying Boolean expression specifies how a logic signal is formed from the values of other signals. The signal assignment statements in the architecture *Boolean_Equations* of entity *or_and_vhdl* specify how a simulator determines the values of *D* and *E* from *A, B*, and *C*.

```
library ieee;
 use ieee.std_logic_1164.all;

 entity or_and_vhdl  is
    port (A, B, C: in  std_logic; E: out  std_logic);
 end  or_and_vhdl;

 architecture Boolean_Equations of or_and_vhdl is
     signal D: std_logic;
 begin
    D <= A  or B;
    E <= C and D;
 end Boolean_Equations ;
```

The reference to **library** ieee in this example indicates that the data types are specified by the standard IEEE library. The data type *std_logic* is a type defined in the language standard *ieee.std_logic_1164*, but it is not part of the VHDL language standard. We will

[9] In a VHDL signal assignment statement, <= denotes a *signal assignment* operator; "**or**" and "**and**" are logical operators. The syntax of various forms of a signal assignment statement is given in Chapter 4.

discuss it in more detail later. Note, though, that every file containing a VHDL model that references the data types in *ieee.std_logic_1164* must contain the library/use statements referencing *ieee.std_logic_1164*.

Practice Exercise 3.13 – VHDL

Write a signal assignment statement that implements the logic diagram in Fig. PE3.13.

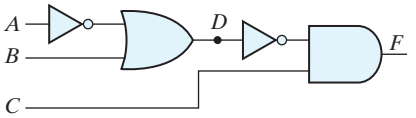


FIGURE PE 3.13

Answer: $F <=$ (**not**((**not** A) **or** B)) **and** C;

Structural (Gate-Level) Modeling

Example 3.1 constructed Verilog and VHDL models based on the Boolean equations implied by the logic diagram of a circuit. Another approach is to use language constructs directly to form a structural model of a circuit. Structural models describe how a circuit is composed of other interconnected elements, such as logic gates or functional blocks.

Verilog

Verilog has a family of built-in structural objects, called *primitives*, that enable direct modeling of combinational logic. For our purposes, the important keyword names of the Verilog primitives are **and**, **nand**, **or**, **nor**, **xor**, **xnor**, **buf**, **not**, **bufif0**, **bufif1**, **notif0**, and **notif1**. They will be described briefly here.[10] Most Verilog primitives are multiple-input primitives—they automatically accommodate two or more inputs. Thus, the same keyword denotes a two-input or a five-input gate.

Verilog Example 3.3 (Structural Modeling with Primitives)

A structural model of the circuit in Fig. 3.37, *and_or_prop_delay*, is specified by a list of (predefined) *primitive* gates, each identified by a descriptive keyword (i.e., **and, not, or**). The circuit has one internal connection, between gates *G1* and *G3*. The gates are connected by *w1*, which is declared with the keyword **wire**. The elements of the list are referred to as *instantiations* of a gate, each of which is referred to as a *gate instance*, or *primitive* instance. Each *gate instance* consists of a primitive name, an optional instance name (such as *G1, G2*, and so on) followed by a list of comma-separated gate output and inputs and enclosed within parentheses. A rule of the language is that the output of

[10] Also see Section 4.12.

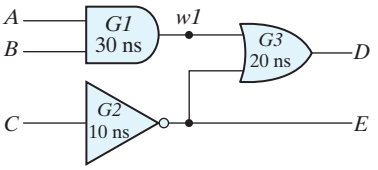


FIGURE 3.37
Schematic for *and_or_prop_delay*

a primitive gate must be listed first, followed by the inputs. For example, the OR gate of the schematic is represented by the **or** primitive, has instance name *G3*, and has output *D* and inputs *w1* and *E*. (*Note*: Although the output of a primitive must be listed first, the inputs and outputs of a module may be listed in any order.) The module description ends with the keyword **endmodule**. Each statement must be terminated with a semicolon, but a semicolon after **endmodule** is not required.

The gates within a module may be listed in any order. They operate concurrently in simulation. A signal can affect simultaneously all of the gates to which it is connected as an input. Each affected gate independently determines and schedules events[11] for its output.

Verilog primitives have built-in logic determining their behavior. The optional user-specified propagation delays (e.g., 30 ns) determine the time interval between a change to the input signal of a gate and the effect apparent at the output of the gate, that is, the model reflects the fact that the input/output time response of a physical logic gate is not instantaneous.[12]

```
'timescale 1 ns / 1 ps                    // time units / resolution
module and_or_prop_delay (
    input A, B, C;
    output D, E);
);
    wire w1;

    and G1 #30 (w1, A, B);          // Prop delay: 30 ns
    not G2 #10 (E, C);              // Prop delay: 10 ns
    or G3 #20 (D, w1, E);           // Prop delay: 20 ns
endmodule
```

It is important to understand the distinction between the terms *declaration* and *instantiation*. The declaration of a Verilog module specifies the input–output behavior of the hardware that it represents. Predefined primitives are not declared, because their

[11] The term *event* denotes a change in the logic value of a signal.

[12] The timescale directive ('timescale 1 ns / 1 ps) specifies that the numerical values in the model are to be interpreted in units of nanoseconds, with a precision of picoseconds. This information would be used by a simulator.

definition is specified by the language and is not subject to change by the user. Primitives are used (i.e., instantiated), just as gates are used to populate a printed circuit board. We'll see that once a module has been declared, it may be used (instantiated) within another module in the design. The sequential ordering of the statements instantiating gates in the model has no significance and does not specify a sequence of computations.

A Verilog model is a *descriptive* model. *and_or_prop_delay* specifies which primitives form the circuit and how they are connected. The input–output behavior of the circuit is implicitly specified by the description because the behavior of each logic gate is predefined. Thus, a Verilog HDL-based model can be used to simulate the circuit that it represents. The gates within a module operate concurrently in simulation. It is essential to realize that the statements that instantiate the gates in an architecture are not a recipe for computing the value of some signal, as they might be in an ordinary programming language that prescribes sequential execution of statements. The order in which gates are referenced during simulation depends on the activity of the signals in the design, not on the order in which the statements are listed. An event (i.e., transition) of a signal activates all of the gates to which it is connected as an input. Each affected gate is evaluated to determine and schedule an event for its output. Physical hardware behaves the same way.

Gate Delays

All physical circuits exhibit a propagation delay between the transition of an input and a resulting transition of an output. When an HDL model of a circuit is simulated, it is sometimes necessary to specify the amount of delay from the input to the output of its gates. In Verilog, the propagation delay of a gate is specified in terms of *time units* and by the symbol #. The numbers associated with time delays in Verilog are dimensionless. The association of a time unit with physical time is made with the '**timescale** compiler directive. (Compiler directives start with the (') back quote, or grave accent, symbol.) Such a directive is specified before the declaration of a module and applies to all numerical values of time in the code that follows. An example of a timescale directive is

```
`timescale 1 ns/100 ps
```

The first number specifies the unit of measurement for time delays. The second number specifies the precision for which the delays are rounded off, in this case to 0.1 ns. If no timescale is specified, a simulator may display dimensionless values or default to a certain time unit, usually 1 ns ($= 10^{-9}$ s). Our examples will use only the default time unit.

The simple circuit in Fig. 3.37 has propagation delays specified for each gate. The **and, or,** and **not** gates have a time delay of 30, 20, and 10 ns, respectively. If the circuit is simulated and the inputs change from $A, B, C = 0$, to $A, B, C = 1$, the outputs change as shown in Table 3.5 (calculated by hand or generated by a simulator). The output of the inverter at E changes from 1 to 0 after a 10 ns delay. The output of the AND gate at *w1* changes from 0 to 1 after a 30 ns delay. The output of the OR gate at D changes from 1 to 0 at $t = 30$ ns and then changes back to 1 at $t = 50$ ns. In both cases, the change in the output of the OR gate results from a change in its inputs 20 ns earlier. It is clear from this result that although

Table 3.5
Output of Gates after Delay

	Time Units (ns)	Input	Output		
		A B C	*E*	*w1*	*D*
Initial	—	0 0 0	1	0	1
Change	—	1 1 1	1	0	1
	10	1 1 1	0	0	1
	20	1 1 1	0	0	1
	30	1 1 1	0	1	0
	40	1 1 1	0	1	0
	50	1 1 1	0	1	1

output *D* eventually returns to a final value of 1 after the input changes, the gate delays produce a negative spike that lasts 20 ns before the final value is reached.

To simulate a circuit with an HDL, it is necessary to apply inputs to the circuit so that the simulator will generate an output response. An HDL description that provides the stimulus to a design is called a *test bench.* Test benches are explained in more detail at the end of Section 4.12. Here, we demonstrate the procedure without dwelling on too many details.

A test bench for *and_or_prop_delay* is given below:

```
// Test bench for and_or_prop_delay

module t_and_or_prop_delay;
   wire       D, E;
   reg        A, B, C;
   and_or_prop_delay M_UUT (A, B, C, D, E);    // Instance name (M_UUT) is required

   initial begin
    A = 1'b0; B = 1'b0; C = 1'b0;
    #100 A = 1'b1; B = 1'b1; C = 1'b1;
   end

   initial #200 $finish;
endmodule
```

In its simplest form, a test bench module contains a signal generator and an instantiation of the model that is to be verified. Note that the test bench (*t_and_or_prop_delay*) has no input or output ports, because it does not interact with its environment. In general, we prefer to name the test bench with the prefix *t_* prepended to the name of the module that is to be tested by the test bench, but that choice is left to the designer. Within the test bench, the stimulus signals that are to be the inputs to the circuit are declared with keyword **reg** and the signals that are connected to the outputs of the circuit are declared with the keyword **wire**. The module *and_or_prop_delay* is *instantiated* with the user-chosen instance name M_UUT (module unit under test). Every

instantiation of a module must include a unique *instance name*. Note that using a test bench is similar to testing actual hardware by attaching signal generators to the inputs of a circuit and attaching probes (wires) to the outputs of the circuit.

Hardware signal generators are not used to verify an HDL model. Instead, the entire simulation exercise is done with software models executing on a digital computer under the direction of an HDL simulator. The waveforms of the input signals are abstractly modeled (generated) by Verilog statements specifying waveform values and transitions. The **initial** keyword is accompanied by a set of statements that begin executing when the simulation is initialized; the signal activity associated with **initial** terminates execution when the last statement has finished executing. The **initial** statements are commonly used to describe waveforms in a test bench. The set of statements to be executed is called a *block statement* and consists of several statements enclosed by the keywords **begin** and **end**. They are executed sequentially, in the order in which they are listed.

The action specified by an **initial** block begins when the simulation is launched, and the statements are executed in sequence, subject to time delays (e.g., #100), left to right, from top to bottom, by a simulator in order to provide the input to the circuit. Initially, $A, B, C = 0$. (A, B, and C are each set to 1'b0, which signifies one binary digit with a value of 0.) After 100 ns, the inputs change to $A, B, C = 1$. After another 100 ns, the simulation terminates at time 200 ns. A second **initial** statement uses the **$finish** *system task* to terminate the simulation. If a statement is preceded by a delay value (e.g., #100), the simulator postpones executing the statement, and any following statements, until the specified time delay has elapsed. The timing diagram of waveforms that result from the simulation of *and_or_prop_delay* is shown in Fig. 3.38. The total simulation generates waveforms over an interval of 200 ns. The inputs A, B, and C change from 0 to 1 at t = 100 ns. Because of propagation delays, output E is unknown for the first 10 ns (denoted by shading), and output D is unknown for the first 30 ns. Output E goes from 1 to 0 at 110 ns. Output D goes from 1 to 0 at 130 ns and back to 1 at 150 ns.

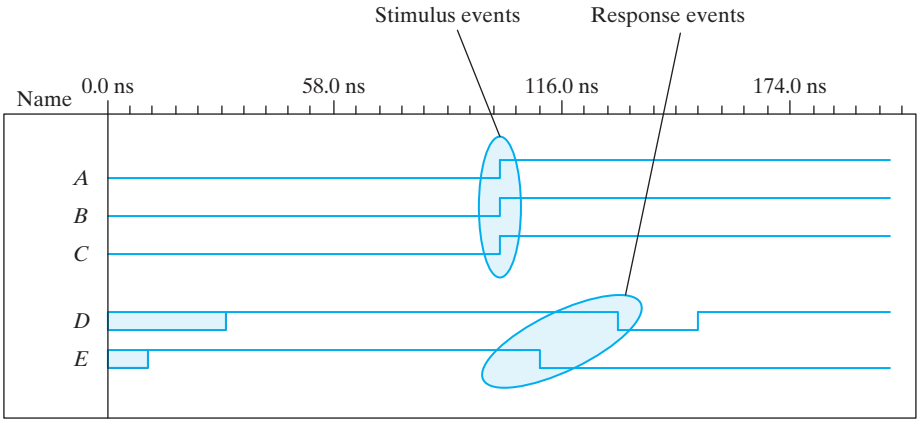


FIGURE 3.38
Simulation waveforms of *and_or_prop_delay*

VHDL Example 3.2 (Structural Modeling with Components)

VHDL does not have built-in combinational logic elements corresponding to logic gates; instead, the design units for structural modeling have to be built as "user-defined *components*," which are modeled with the built-in language operators (the built-in binary logic operators are **and**, **or**, **nand**, **nor**, **xor**, and **xnor**). Once built, components can be used to build more complex structural models. Thus, structural modeling in VHDL is an indirect process of building components before building a structure using components.

The schematic shown in Fig. 3.39 is a structural description, that is, a connection of logic gates. To write a structural model of the circuit in VHDL code, we first build components *and2_gate*, *or2_gate*, and *inv_gate* observing the following syntax template and including the indicated propagation delays (optionally) in signal assignment statements.[13]

```
component component_name
    port (signal_names : mode signal_type;
          signal_names : mode signal_type;
                       . . .
          signal_names : mode signal_type);
end component;

-- Model for 2-input and-gate component[14]

library ieee;
use ieee.std_logic_1164.all;

entity and2_gate is
    port (A, B: in std_logic; w1: out  std_logic);
end and2_gate;
```

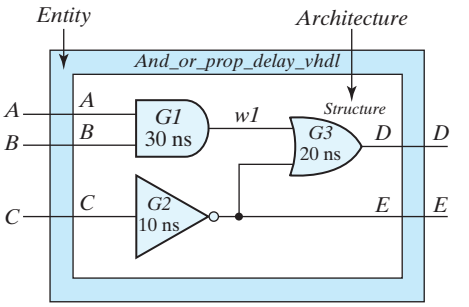


FIGURE 3.39
Entity-architecture for a structural model of *and_or_prop_delay_vhdl*

[13] Propagation delay is optional in a signal assignment statement.

[14] A single line comment in VHDL is denoted by --.

```
architecture Boolean_Operator of and2_gate is
begin
    w1 <= A and B after 30 ns;   -- Logic operator with propagation delay
end architecture Boolean_Operator;

-- Model for 2-input or-gate component

library ieee;
use ieee.std_logic_1164.all;

entity or2_gate is
    port (w1, E: in  std_logic; D: out std_logic);
end or2_gate;

architecture Boolean_Operator of  or2_gate is
begin
    D <= w1 or E after 20 ns;  -- Logic operator
end architecture Boolean_Operator;

-- Model for inverter component

library ieee;
use ieee.std_logic_1164.all;

entity inv_gate is
    port (A: in std_logic; B: out std_logic);
end inv_gate;

architecture Boolean_Operator of inv_gate is
begin
    B <=  not A after 10 ns;
end architecture Boolean_Operator;
```

Next, we declare an entity-architecture pair for *and_or_prop_delay_vhdl*. The components to be used are listed with their ports; the internal signal $w1$ is declared; and then components are instantiated to create a structural model.[15] Each instantiation has a unique instance name ($G1$, $G2$, $G3$).

```
library ieee;
use ieee.std_logic_1164.all;

entity and_or_prop_delay_vhdl is
    port (A, B, C: in std_logic; D: out std_logic; E: buffer std_logic);
end Simple_Circuit_vhdl;

architecture Structure of and_or_prop_delay_is
   component and2_gate       -- Component declaration
    port (w1: out  std_logic; A, B: in std_logic);
   end component;
```

[15] A component is defined only once within an architecture, but it may be instantiated multiple times.

```
    component or2_gate      -- Component declaration
     port (w1:  out  std_logic; A, B:  in  std_logic);
    and component;

    component inv_gate      -- Component declaration
     port (B: out  std_logic; A: in  std_logic);
    end component;

signal w1: std_logic;

begin                           -- Component instantiations
        G1: and2_gate    port map (w1, A, B);
        G2: inv_gate     port map (E, C);
        G3: or2_gate     port map (D, w1, E);

end architecture Structure;
```

Note: The port maps of the components in the preceding example associate by position the names of the ports in the instantiated component with the ports of the declared component. That mechanism is error-prone because it is easy to mistakenly write port names out of order. An alternative style, that is safer and essential when there are many signals in a port, associates the signals of a port's elements by name, in any order, that is, the *formal* names of the ports are associated with the *actual* names of the ports.[16] For example, the gates of *Structure* can be instantiated as follows:

```
G1: and2_gate    port map (B => B w1 => w1, A => A,);
G2: inv_gate     port map (E => E, C => C);
G3: or2_gate     port map (E => E, D => D, w1 => w1);
```

In summary, the process of creating a structural model (1) creates components, (2) declares the components within the top-level architecture, including their ports, (3) instantiates the components and (4) defines port maps making the interconnections of the components forming the structure.

A VHDL structural model is verbose compared to a Verilog model having the same functionality. The process of creating a structural VHDL model is indirect, and requires more coding effort than modeling with the language operators. Nonetheless, the simulated behavior of *and_or_prop_delay_vhdl* is identical to that of the corresponding Verilog model, as shown in Fig. 3.38.

VHDL Packages, Libraries, and Logic Systems

The VHDL mechanisms of libraries and packages promote efficient code, reduced verbosity, and sharing among members of a design team. *A package* provides a repository

[16] The association syntax is *formal_name => actual_name*. In this example the formal and actual names are identical. In general, the formal name is defined when the component is declared; the actual name is defined by the instantiation. A formal name may be associated with multiple actual names when a component is instantiated multiple times.

for declarations that may be common to several design units. A package may have an optional body, which could contain declarations of components, as well as functions, and procedures supporting behavioral models.

A *package statement* has the syntax:

```
package package_name is
    [type_declarations]
    [signal_declarations]
    [constant_declarations]
    [component declarations]
    [function_declarations]
    [procedure_declarations]
end package package_name;

package body package_name is
    [constant_declarations]
    [function_definitions]
    [procedure_definitions]
end [package body][package_name];
```

Signals declared in a package are global signals—they can be referenced by any design entity that use the package.

The package *ieee_std_logic_1164* is not part of the VHDL language. This package defines a 9-valued logic system, which is widely used in industry to model and simulate circuits, especially those based on CMOS technology. The symbols of the standard logic values are given in Table 3.6, which specifies logic values that models may assign to signals in a simulation. Of these, the four values 0, 1, X, and Z are widely used, with X representing a signal whose value is ambiguous, possibly because there are multiple drivers, and Z representing a high impedance condition, as happens if a terminal of a device is unconnected and floating. The don't care value (-) allows a synthesis tool to choose a signal assignment to more efficiently simplify Boolean logic. Weak values are used in modeling logic circuits at the CMOS transistor level and will not be considered in this text.

Table 3.6
Logic Symbols of the IEEE_std_logic_1164 Package

'U'	Uninitialized
'X'	Strong drive, unknown logic value
'0'	Strong drive, logic 0
'1'	Strong drive, logic 1
'Z'	High impedance
'W'	Weak drive, unknown logic value
'L'	Weak drive, logic 0
'H'	Weak drive, logic 1
'-'	Don't care

If components are declared within a package they may be referenced by the architecture of any entity whose declaration is preceded by the related package statement. This practice eliminates, for example, the need to have multiple declarations of the gates within the entities of a design. Each entity that uses a gate that is declared in a package needs only to instantiate it within its architecture.

A VHDL *library* is a more general repository containing packages and the compiled models declared in the packages.[17] The preceding examples illustrate how the contents of a package may be referenced—note that each entity is preceded with the following pair of statements:

```
library ieee;
use ieee.std_logic_1164.all;
```

The first statement identifies a specific library (ieee); the second statement identifies within that library, by a "**use**" clause, a package to be compiled in its entirety.[18]

Packages and libraries simplify structural design because components that are held within a package do not have to be re-declared before they are instantiated within an architecture.

3.10 TRUTH TABLES IN HDLs

The preceding examples have illustrated HDL models of logic circuits described by Boolean equations and by logic gates. Combinational logic may also be described by a truth table. Not all HDLs support truth table descriptions of digital logic.

Verilog—User-Defined Primitives

The logic gates used in Verilog descriptions with keywords **and, or, etc., are** defined by the system and are referred to as *system primitives. (Caution: Other languages may use these words differently.)* The user can create additional primitives by defining them in tabular form. These types of circuits are referred to as *user-defined primitives* (UDPs). One way of specifying a digital circuit in tabular form is by means of a truth table. UDP descriptions do not use the keyword pair **module** . . . **endmodule**. Instead, they are declared with the keyword pair **primitive** . . . **endprimitive**.

Verilog Example 3.4 defines a UDP with a truth table. It proceeds according to the following general rules:

- A UDP is declared with the keyword **primitive**, followed by a name and port list.
- There can be only one output, and it must be listed first in the port list and declared with keyword **output**.

---

[17] The compilers in VHDL-based design tools automatically generate a design-specific library named *work*, which serves as a repository for the compiled files of a design project.

[18] Replacing *.all* by *.all.and2_gate* would restrict access to a particular model (*and2_gate*) within the package.

- There can be any number of inputs. The order in which they are listed in the **input** declaration must conform to the order in which they are given values in the table that follows.
- The truth table is enclosed within the keywords **table** and **endtable**.
- The values of the inputs are listed in order, ending with a colon (:). The output is always the last entry in a row and is followed by a semicolon (;).
- The declaration of a UDP ends with the keyword **endprimitive**.

Verilog Example 3.4 (User-Defined Primitive)

```
// Verilog model: User-defined Primitive
primitive UDP_02467 (D, A, B, C);
 output D;
 input  A, B, C;
//Truth table for D = f (A, B, C) = ∑(0, 2, 4, 6, 7);
 table
//A   B   C   :   D     // Column header comment
 0   0   0   :   1;
 0   0   1   :   0;
 0   1   0   :   1;
 0   1   1   :   0;
 1   0   0   :   1;
 1   0   1   :   0;
 1   1   0   :   1;
 1   1   1   :   1;
 endtable
endprimitive
// Instantiate primitive
// Verilog model: Circuit instantiation of Circuit_UDP_02467
module Circuit_with_UDP_02467 (e, f, a, b, c, d);
 output       e, f;
 input        a, b, c, d;
 UDP_02467    (e, a, b, c);
 and          (f, e, d);  // Optional gate instance name omitted
endmodule
```

Note that the variables listed at the top of the table are part of a comment and are shown only for clarity. The system recognizes the variables by the order in which they are listed in the input declaration. A user-defined primitive can be instantiated in the construction of other modules (digital circuits), just as the system primitives are used. For example, the declaration

```
Circuit_with_UDP_02467(E,F,A,B,C,D);
```

will produce a circuit that implements the hardware shown in Fig. 3.40.

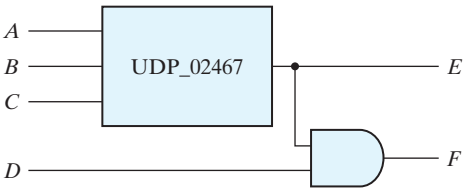


FIGURE 3.40
Schematic for *Circuit with_UDP_02467*

Although Verilog HDL uses this kind of description for UDPs only, other HDLs and computer-aided design (CAD) systems use other procedures to specify digital circuits in tabular form. The tables can be processed by CAD software to derive an efficient gate structure of the design. None of Verilog's predefined primitives describes sequential logic. The model of a sequential UDP requires that its output be declared as a **reg** data type, and that a column be added to the truth table to describe the next state. So the columns are organized as inputs : state : next state.

In this section, we introduced the HDLs and presented simple examples to illustrate alternatives for modeling combinational logic. A more detailed presentation of modeling with HDLs can be found in the next chapter. The reader familiar with combinational circuits can go directly to Section 4.12 to continue with this subject.

VHDL—Truth Tables

VHDL does not support truth tables directly. Instead, a truth table has to be converted into a set of Boolean equations, which can be described by signal assignment statements.

PROBLEMS

(Answers to problems marked with * appear at the end of the text.)

3.1* Simplify the following Boolean functions, using three-variable K-maps:
(a) $F(x, y, z) = \Sigma(0, 2, 3, 7)$ (b) $F(x, y, z) = \Sigma(2, 3, 5, 6, 7)$
(c) $F(x, y, z) = \Sigma(0, 1, 2, 4, 6)$ (d) $F(x, y, z) = \Sigma(1, 3, 5, 7)$

3.2 Simplify the following Boolean functions, using three-variable K-maps:
(a)* $F(x, y, z) = \Sigma(2, 3, 4, 5, 7)$ (b)* $F(x, y, z) = \Sigma(0, 1, 4, 5, 6, 7)$
(c) $F(x, y, z) = \Sigma(0, 1, 2, 4, 5, 6)$ (d) $F(x, y, z) = \Sigma(1, 2, 3, 5, 6, 7)$
(e) $F(x, y, z) = \Sigma(1, 3, 5, 7)$ (f) $F(x, y, z) = \Sigma(0, 1, 2, 3, 5, 7)$

3.3* Simplify the following Boolean expressions, using three-variable K-maps:
(a)* $F(x, y, z) = x'y'z + xyz' + x'yz + xy$ (b)* $F(x, y, z) = xyz' + y'z' + xz$
(c)* $F(x, y, z) = x'yz + xy' + yz'$ (d) $F(x, y, z) = xz' + x'z + x'y + xy'$